The time-modulated LISA Galactic foreground in the wavelet domain: data gaps and a needlet description of the sky

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Abstract

The Galactic foreground (GF) of unresolved white-dwarf binaries is the loudest stochastic signal in LISA, and its power rises and falls as the constellation's antenna pattern sweeps across the Galaxy. This work develops a wavelet-domain analysis of that modulation. Part I reviews the time–frequency ("chunked") Fourier analysis of Buscicchio, Pozzoli et al. (2025) and the envelope model of Buscicchio, Klein et al. (2025). Part II builds the Wilson–Daubechies–Meyer (WDM) alternative from scratch, models the modulation with a variable number of Gaussian atoms sampled by reversible-jump MCMC, and shows that the wavelet Gamma likelihood is Buscicchio, Pozzoli et al. (2025), Eq. (9) written in another domain; the derivation exposes a missing factor of two in the degrees of freedom of that equation. Part III shows how data gaps are handled in the WDM domain: the damaged pixels are left out of the likelihood, which is exact because the pixels are independent. Part IV introduces spherical needlets, a basis on the sky that is local in both position and multipole, and uses it to understand which features of a patchy Galaxy LISA can see. Part V presents five injection studies on 30 days of simulated data: recovery of an atom modulation, a comparison with the chunked pipeline on the true envelope, a 1-day and a 10-day gap, and a Galaxy with four bright patches of excess power. Every equation taken from a paper is given with a link to it.

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Part I
The chunked Fourier analysis

1 The idea

The Galactic foreground is the incoherent superposition of millions of unresolved double white-dwarf binaries in the Milky Way. Its power spectrum is not constant in time. The Galaxy is concentrated towards the Galactic centre, and LISA's antenna pattern sweeps across it as the constellation orbits the Sun and rotates. The power in each TDI channel therefore rises and falls with a one-year period, twice per year. A process whose statistics vary periodically in time is called *cyclostationary*.

The modulation is *slow*: it changes appreciably over weeks to months. This suggests the following strategy:

1. split the data stream into N_c non-overlapping *chunks* of duration T (one to a few weeks);
2. inside each chunk, treat the data as *stationary*, with a PSD whose galactic part is scaled by the modulation averaged over that chunk;
3. treat different chunks as independent, so the total log-likelihood is a sum over chunks.

The same construction handles *data gaps*. Chunks are simply placed in the stretches of data between gaps, and no special treatment of the gaps is needed.

The total log-likelihood is (Buscicchio, Pozzoli et al. (2025), Eq. (1))

$$\log \mathcal{L}(\boldsymbol{d} \mid \theta) = \sum_{c=1}^{N_c} \log \mathcal{L}^{\mathrm{chunk}}(\boldsymbol{d}^c \mid \theta), \tag{1}$$

where $\boldsymbol{d}^c$ is the data of chunk c and θ the full set of model parameters. The rest of this note explains what goes into a single $\log \mathcal{L}^{\mathrm{chunk}}$.

Notation.

$c = 1, \ldots, N_c$	chunk index
$j \in \{A, E\}$	TDI channel
$k = 1, \ldots, n_f$	fine (full-resolution) frequency bin
$m = 1, \ldots, n_g$	coarse (log-spaced) frequency bin
$l = 1, \ldots, N$	time sample within a chunk
$T,\ \mathrm{d}t$	chunk duration, sampling cadence
$N = T/\mathrm{d}t,\ f_s = 1/\mathrm{d}t$	samples per chunk, sampling frequency
$t_S^c,\ t_E^c$	start and end time of chunk c
$S_{jk}^c(\theta)$	model PSD of channel j, chunk c, frequency f_k

Overview. Figure 1 shows the whole pipeline. There are two branches. The *data* branch (left) is executed *once*, with the injected parameters. The *model* branch (right) is executed at *every* step of the sampler, with the trial parameters θ. The two meet in the likelihood.

2 The frequency grid of a chunk

A chunk of duration T sampled at cadence $\mathrm{d}t$ contains $N = T/\mathrm{d}t$ samples. Its discrete Fourier transform lives on the grid

$$f_k = \frac{k}{T}, \qquad \Delta f = \frac{1}{T}, \tag{2}$$

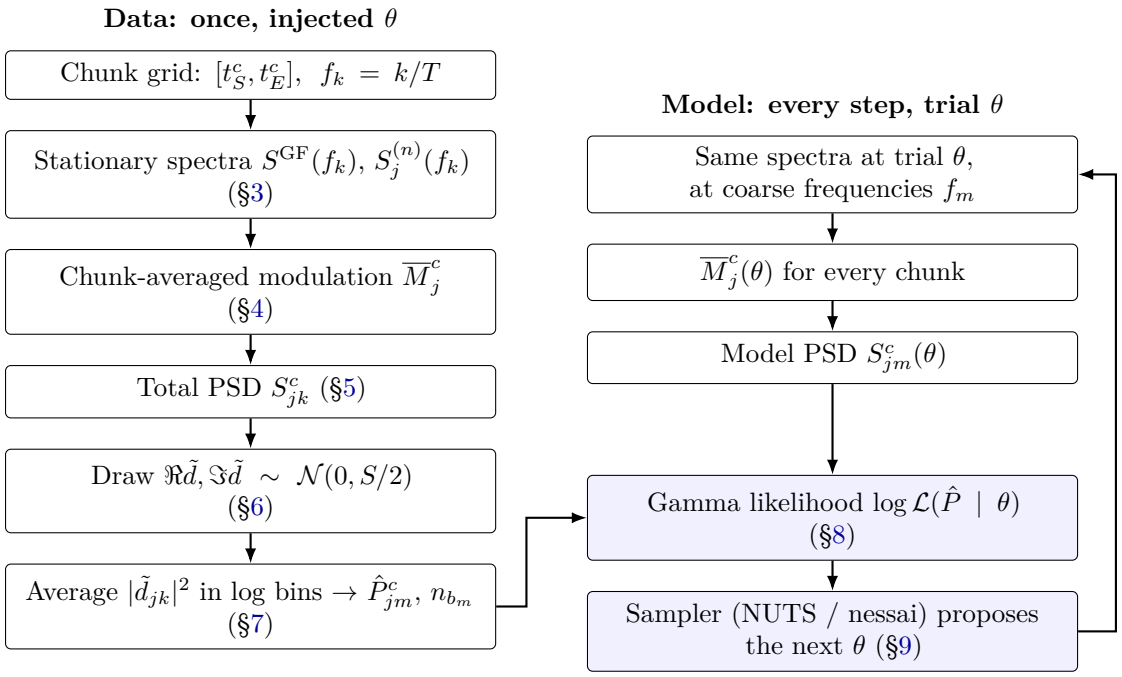


Figure 1: The chunked pipeline. The data are generated once and then frozen. During sampling only the model branch is recomputed: no new random draws and no new coarse graining.

and n_f denotes the number of grid points inside the analysis band $[f_{\min}, f_{\max}]$, i.e. $n_f \simeq (f_{\max} - f_{\min})\,T$. Longer chunks give finer frequency resolution. With equal-duration chunks, every chunk has the *same* grid.

The Fourier amplitudes are normalised so that their squared modulus is a one-sided PSD estimate (Buscicchio, Pozzoli et al. (2025), Eq. (2)):

$$\tilde{d}_{jk}^c = \sqrt{\frac{2}{W f_s}} \sum_{l=1}^{N} w_l\, d_{jl}^c \exp\left[-\frac{2\pi\imath}{N}\, k\, l\right], \qquad W = \sum_{l=1}^{N} |w_l|^2, \tag{3}$$

where w_l is a window function. For data generated directly in the frequency domain (Section 6) no window is needed and $W = N$. With this normalisation $\langle |\tilde{d}_{jk}^c|^2 \rangle = S_{jk}^c$.

Full-resolution (Whittle) likelihood. If the data in a chunk are Gaussian and stationary, the Fourier amplitudes at different frequencies are independent, and the chunk likelihood is the Whittle likelihood (Buscicchio, Pozzoli et al. (2025), Eq. (3)):

$$\log \mathcal{L}_W^{\mathrm{chunk}}(\tilde{\boldsymbol{d}}^c \mid \theta) = -\frac{1}{2} \sum_{j \in \{A,E\}} \sum_{k=1}^{n_f} \left(\frac{|\tilde{d}_{jk}^c|^2}{S_{jk}^c(\theta)} + \log S_{jk}^c(\theta) + \log 2\pi \right). \tag{4}$$

It is exact but expensive. At every sampler step it costs $N_c \times 2 \times n_f$ terms. For example, a two-week chunk has $\Delta f \approx 0.83\,\mu$Hz, so a band of tens of mHz holds tens of thousands of fine bins per channel and per chunk. This motivates the coarse-grained alternative of Section 7.

3 Stationary spectra: Galactic foreground shape and instrument noise

Galactic foreground spectrum. The spectral shape of the GF is the phenomenological model of Karnesis et al. (2021) (Buscicchio, Pozzoli et al. (2025), Eq. (11)):

$$S^{\mathrm{GF}}(f) = \frac{A}{2} f^{-7/3} \exp\left[-\left(\frac{f}{f_1}\right)^{\alpha}\right] \left[1 + \tanh\left(\frac{f_{\mathrm{knee}} - f}{f_2}\right)\right], \tag{5}$$

with amplitude A, slope parameter α, and characteristic frequencies f_1, f_{knee}, f_2. The $f^{-7/3}$ is the inspiral spectrum of a population of circular binaries. The tanh produces the high-frequency drop above f_{knee}, where binaries become sparse enough to be resolved and subtracted.

Instrument noise. Two noise sources remain after TDI: test-mass (TM) acceleration noise and optical-metrology-system (OMS) noise (Buscicchio, Pozzoli et al. (2025), Eq. (16) and Buscicchio, Pozzoli et al. (2025), Eq. (17)):

$$S^{n}_{\mathrm{tm}}(f) = \left(A_{\mathrm{n}} \times 10^{-15}\,\mathrm{m\,s^{-2}\,Hz^{-1/2}}\right)^2 \frac{1}{(2\pi c f)^2} \left[1 + \left(\frac{0.4\,\mathrm{mHz}}{f}\right)^2\right] \left[1 + \left(\frac{f}{8\,\mathrm{mHz}}\right)^4\right], \tag{6}$$

$$S^{n}_{\mathrm{oms}}(f) = \left(P \times 10^{-12}\,\mathrm{m\,Hz^{-1/2}}\right)^2 \left(\frac{2\pi f}{c}\right)^2 \left[1 + \left(\frac{2\,\mathrm{mHz}}{f}\right)^4\right], \tag{7}$$

where c is the speed of light. (The paper calls the TM amplitude A; we write A_{n} to avoid a clash with the GF amplitude.) Both components are projected onto the A and E channels with the TDI transfer functions of Nam et al. (2023), giving the channel noise PSD $S^{(n)}_j(f)$. The injected values are $A_{\mathrm{n}} = 2.4$ and $P = 7.9$.

Why these do not depend on the chunk

$S^{\mathrm{GF}}(f)$ and $S^{(n)}_j(f)$ depend only on frequency. They are evaluated on the chunk grid f_k, which is the same for every chunk. All the time dependence of the model comes from the modulation factor of Section 4.

Remark: the GF shape evolves with observation time

In Buscicchio, Pozzoli et al. (2025), f_1 and f_{knee} are not constants but functions of the total observation time (Buscicchio, Pozzoli et al. (2025), Eq. (12) and Buscicchio, Pozzoli et al. (2025), Eq. (13)):

$$\log_{10} f_1 = a_1 \log_{10}(T_{\mathrm{obs}}) + b_1, \qquad \log_{10} f_{\mathrm{knee}} = a_{\mathrm{knee}} \log_{10}(T_{\mathrm{obs}}) + b_{\mathrm{knee}}, \tag{8}$$

with calibrated values $a_1 = -0.25$, $b_1 = -2.7$, $a_{\mathrm{knee}} = -0.27$, $b_{\mathrm{knee}} = -2.47$. Both slopes are negative. The longer LISA observes, the more binaries are resolved and subtracted, and the cutoff moves to lower frequency.

To mimic a global fit, where the foreground estimate is continuously updated as individual sources are added or removed, the paper simulates the chunks *sequentially*. Each new chunk's spectrum uses the observation time accumulated up to that chunk. The GF spectrum then drifts slowly to lower frequency from chunk to chunk, on top of the up-and-down drift caused by the modulation. If T_{obs} is held fixed, as in the rest of this note, the shape is the same for all chunks.

4 The orbital modulation

The modulation is derived in Buscicchio, Klein et al. (2025) (main text Sec. 6 and Appendix A) in the *low-frequency approximation*. In this limit, A and E behave as two independent effective Michelson interferometers whose orientation follows LISA's orbit.

4.1 Step 1: many sources at a single sky location

Consider a large number of binaries at the same sky position: ecliptic latitude and longitude (b_S, l_S). Their inclinations ι and polarisation angles ψ are random. The squared response is averaged over the gravitational-wave period and over these angles (Buscicchio, Klein et al. (2025), Eq. (27)):

$$\langle h_{A,E}\rangle^2(t, b_S, l_S) = \frac{1}{T}\int_0^T \mathrm{d}t' \int_{-1}^{1} \mathrm{d}\cos\iota \int_0^{2\pi} \mathrm{d}\psi \; h^2_{A,E}. \tag{9}$$

Here T is the *period of the gravitational wave* (minutes to hours). The average removes the fast oscillation of the signal and keeps only its slowly varying amplitude. The result still depends on time t, but only through LISA's annual motion.

The averaged envelope is most conveniently written through the sum and difference of the two channels. Both are short Fourier series in the orbital phase (Buscicchio, Klein et al. (2025), Eq. (A.35)–(A.40)):

$$h^2_S \equiv \langle h_A\rangle^2 + \langle h_E\rangle^2 = \sum_{n=0}^{4} h^2_{S;(n)}(b_S)\,\cos\left(n\,\Delta l_L\right), \tag{10}$$

$$h^2_D \equiv \langle h_A\rangle^2 - \langle h_E\rangle^2 = \sum_{n=0}^{8} h^2_{D;(n)}(b_S)\,\cos\left(n\,\Delta l_L + 4\bar{l}_S\right), \tag{11}$$

with $\Delta l_L = \Omega t - l_S$, $\bar{l}_S = l_S + \pi/12$, $\Omega = 2\pi/\mathrm{yr}$. The coefficients $h^2_{S;(n)}$, $h^2_{D;(n)}$ are polynomials in $\sin b_S$ and $\cos b_S$ (Buscicchio, Klein et al. (2025), Eq. (A.41)–(A.54)). For example, $h^2_{S;(0)} = \frac{9}{320}\left(328 + 152\cos^2 b_S - 37\cos^4 b_S\right)$. So the sum $A^2 + E^2$ contains harmonics up to 4Ω, and the difference up to 8Ω.

4.2 Step 2: a model for the sky distribution

The Galaxy is not a point. Its unresolved binaries are modelled with a (deliberately simple, and admittedly unrealistic) bivariate Gaussian on the sky (Buscicchio, Klein et al. (2025), Eq. (A.55)–(A.57)):

$$p(b_S, l_S) = \frac{1}{2\pi\sigma_1\sigma_2}\exp\left[-\frac{1}{2}\,\boldsymbol{\theta}^T\mathbb{M}\,\boldsymbol{\theta}\right], \qquad \boldsymbol{\theta} = \begin{pmatrix} b_S - b_M \\ l_S - l_M \end{pmatrix}, \tag{12}$$

$$\mathbb{M} = \begin{pmatrix} \cos\delta & \sin\delta \\ -\sin\delta & \cos\delta \end{pmatrix}\begin{pmatrix} \sigma_1^{-2} & 0 \\ 0 & \sigma_2^{-2} \end{pmatrix}\begin{pmatrix} \cos\delta & -\sin\delta \\ \sin\delta & \cos\delta \end{pmatrix}. \tag{13}$$

It is centred on the Galactic centre (b_M, l_M), has standard deviations $\sigma_{1,2}$ along its principal axes, and is rotated by an angle δ relative to the ecliptic.

4.3 Step 3: integrate over the sky

The envelope of the whole Galaxy is the single-location envelope weighted by p and integrated over the sky (Buscicchio, Klein et al. (2025), Eq. (A.58)):

$$H^2_{S,D}(t) = \int_{\mathbb{R}} \cos b_S\,\mathrm{d}b_S \int_{\mathbb{R}} \mathrm{d}l_S \; p(b_S, l_S)\, h^2_{S,D}. \tag{14}$$

For simplicity the integral runs over the whole plane rather than over the sphere. This is a good approximation as long as p falls off quickly. The integral can be done in closed form because every term of $h^2_{S,D}$ is a product of sines and cosines of b_S and l_S. The integral of $e^{inb_S}e^{iml_S}$ against a Gaussian is again a Gaussian factor (its characteristic function, Buscicchio, Klein et al. (2025), Eq. (A.59)). Each harmonic is therefore damped by a factor that depends on the widths and orientation of the Galaxy. With the shorthand (Buscicchio, Klein et al. (2025), Eq. (A.63)–(A.69))

$$\sigma_S^2 = \sigma_1^2 + \sigma_2^2, \quad \sigma_D^2 = \sigma_1^2 - \sigma_2^2, \quad \sigma_c^2 = \sigma_D^2 \cos 2\delta, \quad \sigma_s^2 = \sigma_D^2 \sin 2\delta, \quad \sigma_+^2 = \sigma_S^2 + \sigma_c^2, \tag{15}$$

$$\bar{l}_M = l_M + \frac{\pi}{12}, \qquad \Delta\phi_L = \Omega t - l_M, \tag{16}$$

the result is again a finite Fourier series in $\Delta\phi_L$ (Buscicchio, Klein et al. (2025), Eq. (A.61), (A.62)),

$$H_S^2 = \frac{1}{10240}\sum_{n=0}^{4} H^2_{S;(n)}, \qquad H_D^2 = \frac{1}{10240}\sum_{n=0}^{8} H^2_{D;(n)}, \tag{17}$$

where $H^2_{S;(n)}$ and $H^2_{D;(n)}$ contain the n-th harmonic. The first two terms are (Buscicchio, Klein et al. (2025), Eq. (A.70), (A.71))

$$H^2_{S;(0)} = e^{-\frac{1}{4}\sigma_S^2 - \frac{1}{4}\sigma_c^2}\left(120636\cos b_M + 7614\, e^{-2\sigma_+^2}\cos 3b_M - 666\, e^{-6\sigma_+^2}\cos 5b_M\right), \tag{18}$$

$$\begin{aligned} H^2_{S;(1)} = \sqrt{3}\, e^{-\frac{1}{2}\sigma_S^2}\Big[&- 31392\Big(\cosh\tfrac{\sigma_s^2}{2}\sin b_M\cos\Delta\phi_L + \sinh\tfrac{\sigma_s^2}{2}\cos b_M\sin\Delta\phi_L\Big) \\ &- 32112\, e^{-2\sigma_+^2}\Big(\cosh\tfrac{3\sigma_s^2}{2}\sin 3b_M\cos\Delta\phi_L + \sinh\tfrac{3\sigma_s^2}{2}\cos 3b_M\sin\Delta\phi_L\Big) \\ &- 720\, e^{-6\sigma_+^2}\Big(\cosh\tfrac{5\sigma_s^2}{2}\sin 5b_M\cos\Delta\phi_L + \sinh\tfrac{5\sigma_s^2}{2}\cos 5b_M\sin\Delta\phi_L\Big)\Big]. \end{aligned} \tag{19}$$

The remaining terms (Buscicchio, Klein et al. (2025), Eq. (A.72)–(A.83)) follow the same pattern: a damping exponential, times coefficients in b_M, times cos and sin of $n\Delta\phi_L$ (for H_S) or of $n\Delta\phi_L + 4\bar{l}_M$ (for H_D). Finally, the envelopes of the two channels are (Buscicchio, Klein et al. (2025), Eq. (A.84), (A.85))

$$\boxed{h^2_{A,0}(t) = \tfrac{1}{2}\left(H_S^2 + H_D^2\right), \qquad h^2_{E,0}(t) = \tfrac{1}{2}\left(H_S^2 - H_D^2\right).} \tag{20}$$

These are the squared modulation functions $M_A^2(t)$ and $M_E^2(t)$ used in Buscicchio, Pozzoli et al. (2025). They describe the *power* of the GF in each channel as a function of time. Table 1 lists the parameters under the names used by each source.

Meaning	Klein et al.	Pozzoli et al.	`bahamas`
Galactic-centre latitude	b_M	$\sin\beta$	`lat`
Galactic-centre longitude	l_M	λ	`long`
Variances along the axes	$\sigma_1^2,\ \sigma_2^2$	$\sigma_1^2,\ \sigma_2^2$	`s1`, `s2`
Orientation	δ	$\sin\psi$	`psi` $(= \sin 2\delta)$

Table 1: The five envelope parameters ($\theta_{\mathrm{GF_{II}}}$) as named in Buscicchio, Klein et al. (2025), in Buscicchio, Pozzoli et al. (2025) (which samples the latitude through its sine), and in the code.

A few printed coefficients in App. A of Buscicchio, Klein et al. (2025) contain misprints; the code implements the corrected expressions. Details are in `reports/PAPER_BUGS.md` of the `bahamas` repository.

4.4 Step 4: average over a chunk

Inside a chunk the PSD is assumed stationary, so the modulation is replaced by its average over the chunk (Buscicchio, Pozzoli et al. (2025), Eq. (14)):

$$\overline{M}_j^c = \frac{1}{t_E^c - t_S^c} \int_{t_S^c}^{t_E^c} \mathrm{d}t \; M_j^2(t \mid \theta_{\mathrm{GF_{II}}}). \tag{21}$$

This is a single number per chunk and per channel. Note that this is a *second*, different average. Equation (9) averages over the gravitational-wave period of each binary (fast). Equation (21) averages the resulting slow envelope over the duration of a chunk (weeks).

There is no need to evaluate $M_j^2(t)$ on a fine time grid and integrate it numerically. Equation (20) is a finite sum of terms $\cos(n\Delta\phi_L + \varphi)$ and $\sin(n\Delta\phi_L + \varphi)$, and the average of each term over the chunk is elementary. For $n \geq 1$,

$$\frac{1}{t_E - t_S} \int_{t_S}^{t_E} \mathrm{d}t \; \cos(n\Omega t + \varphi) = \frac{\sin(n\Omega t_E + \varphi) - \sin(n\Omega t_S + \varphi)}{n\,\Omega\,(t_E - t_S)}, \tag{22}$$

and similarly for the sines. Replacing every harmonic by its chunk average gives $\overline{M}_j^c$ *exactly* and at negligible cost. This is what `bahamas` does.

Why the chunk average is a reasonable approximation

The envelope is dominated by low harmonics of the one-year orbit, so it changes little over a week or two. Treating the chunk as stationary with the mean power $\overline{M}_j^c$ discards only the variation of M_j^2 *within* the chunk. The longer the chunk, the larger this error. The shorter the chunk, the fewer frequency bins it has. The chunk duration balances the two.

5 The total PSD of a chunk

Combining the pieces, the PSD of channel j in chunk c is (Buscicchio, Pozzoli et al. (2025), Eq. (10))

$$S_{jk}^c(\theta) = \overline{M}_j^c(\theta_{\mathrm{GF_{II}}})\, S_k^{\mathrm{GF}}(\theta_{\mathrm{GF_I}}) + S_{jk}^{(n)}(\theta_n) + R_{jk}\, S_k^{\mathrm{EF}}(\theta_{\mathrm{EF}}). \tag{23}$$

The first term is the modulated Galactic foreground: a time-independent spectral shape, rescaled chunk by chunk. The second is the stationary instrument noise. The last is an optional stationary, isotropic extragalactic foreground, projected with a chunk-independent response R_{jk}. It is only included when that component is studied, and we set it to zero here. The parameters are grouped as $\theta_{\mathrm{GF_I}} = \{\log_{10} A, \log_{10} f_{\mathrm{knee}}, \log_{10} f_1, \log_{10} f_2\}$ (spectral shape), $\theta_{\mathrm{GF_{II}}}$ (envelope, Table 1) and $\theta_n = \{A_{\mathrm{n}}, P\}$ (noise).

6 Simulating the data

Both the GF and the instrument noise are modelled as zero-mean Gaussian processes. The GF is the sum of very many independent sources, so the central limit theorem applies. The PSD fixes the *variance* at each frequency, not the values of the data. A data set is one random realisation consistent with that variance.

Buscicchio, Pozzoli et al. (2025) generates the data directly in the frequency domain. For each chunk c, channel j and fine frequency f_k, the real and imaginary parts of the Fourier amplitude are drawn independently:

$$\Re\, \tilde{d}_{jk}^c,\ \Im\, \tilde{d}_{jk}^c \ \sim\ \mathcal{N}\Big(0,\ \tfrac{1}{2} S_{jk}^c(\theta_{\mathrm{inj}})\Big)\,, \tag{24}$$

where $\theta_{\rm inj}$ are the injected parameters. $\tilde{d}^c_{jk}$ is a *circular* complex Gaussian with zero mean, and its squared modulus has expectation

$$\langle |\tilde{d}^c_{jk}|^2 \rangle = \langle (\Re \tilde{d})^2 \rangle + \langle (\Im \tilde{d})^2 \rangle = S^c_{jk}. \tag{25}$$

The amplitudes themselves average to zero and carry no information. The information sits in their *squared moduli*, whose mean is the PSD. This is why everything that follows works with $|\tilde{d}|^2$.

Since each chunk is drawn independently, the chunks are independent by construction, and no window is needed ($W = N$ in Equation (3)). The data are generated once and then kept fixed for the whole inference.

7 Coarse graining: averaged periodograms

To reduce the cost of Equation (4), the n_f fine frequencies of each chunk are grouped into $n_g \ll n_f$ *logarithmically spaced* intervals. The interval edges are (Buscicchio, Pozzoli et al. (2025), Eq. (5)–(8))

$$\Delta = \frac{\log_{10} f_{\rm max} - \log_{10} f_{\rm min}}{n_g}, \tag{26}$$

$$\log_{10} f_m = \log_{10} f_{\rm min} + m\,\Delta, \tag{27}$$

$$\mathcal{F}_m = \{\, k \mid f_m < f_k < f_{m+1} \,\}, \tag{28}$$

$$n_{b_m} = |\mathcal{F}_m|. \tag{29}$$

Here $\mathcal{F}_m$ is the set of fine frequencies that fall in the m-th interval, and n_{b_m} is how many there are.[1] Within each interval, the squared amplitudes are replaced by their mean, the *averaged periodogram* (Buscicchio, Pozzoli et al. (2025), Eq. (4)):

$$\hat{P}^c_{jm} = \frac{1}{n_{b_m}} \sum_{k \in \mathcal{F}_m} |\tilde{d}^c_{jk}|^2. \tag{30}$$

After this step each chunk and channel is described by n_g numbers $\hat{P}^c_{jm}$ instead of n_f complex amplitudes. Each $\hat{P}^c_{jm}$ is an estimate of the PSD in its interval, whose scatter shrinks as n_{b_m} grows.

Three practical points:

- Because the intervals have equal *logarithmic* width, they are narrow at low frequency and wide at high frequency. n_{b_m} is therefore not n_f/n_g but grows roughly in proportion to f: a few fine bins near $f_{\rm min}$, hundreds at the high end. Intervals that contain no fine frequency are discarded.

- Averaging assumes that the PSD is nearly constant across an interval. With $n_g = 1000$ (Buscicchio, Pozzoli et al. (2025)) each interval is only a fraction of a percent to about a percent wide in frequency, so this holds wherever the spectrum is smooth.

- The model PSD is evaluated at one representative frequency per interval (in `bahamas`, the midpoint between its first and last fine frequency), giving $S^c_{jm}(\theta)$.

Buscicchio, Pozzoli et al. (2025) uses $f_{\rm min} = 0.1\,\text{mHz}$ and $n_g = 1000$.[2]

[1] Buscicchio, Pozzoli et al. (2025) prints the third line as $\mathcal{F}_k = \{m \mid f_m < f_k < f_{m+1}\}$. The roles of k and m are swapped there; the version above is the intended one.

[2] The paper quotes $f_{\rm max} = 2.9\,\text{mHz}$. The `bahamas` template configuration uses $f_{\rm max} = 0.029\,\text{Hz} = 29\,\text{mHz}$; it is worth confirming which value was used.

8 The Gamma likelihood

8.1 Why a Gamma distribution

Each $|\tilde{d}_{jk}|^2$ is the sum of the squares of two independent zero-mean Gaussians (real and imaginary part), each with variance $S/2$. That is $\frac{S}{2}\chi^2_2$, an exponential distribution with mean S, which is the special case of the Gamma distribution with shape 1 (cf. Buscicchio, Klein et al. (2025), Eq. (21)). The Gamma family is closed under sums: independent Gamma variables with a common scale add up to a Gamma variable whose shape is the sum of the shapes. The averaged periodogram $\hat{P}^c_{jm}$ is the mean of n_{b_m} uncorrelated squared circular Gaussians, so it is Gamma distributed. (By contrast, the amplitudes $\tilde{d}$ themselves are Gaussian, which is why Equation (4) has a Gaussian form.)

Buscicchio, Pozzoli et al. (2025) writes the single-chunk likelihood as (Buscicchio, Pozzoli et al. (2025), Eq. (9))

$$\log \mathcal{L}_G^{\text{chunk}}(\hat{P}^c \mid \theta) = -\sum_{j\in\{A,E\}} \sum_{m=1}^{n_g} \left[\log\Gamma\left(\frac{n_{b_m}}{2}\right) + \frac{n_{b_m}}{2}\log S^c_{jm}(\theta) - \left(\frac{n_{b_m}}{2}-1\right)\log \hat{P}^c_{jm} + \frac{n_{b_m}}{2}\frac{\hat{P}^c_{jm}}{S^c_{jm}(\theta)}\right]. \tag{31}$$

The two roles are worth stating clearly:

- $\hat{P}^c_{jm}$ (with a hat) is the **data**: the averaged periodogram of Equation (30), computed once from the simulated amplitudes.
- $S^c_{jm}(\theta)$ is the **model**: the PSD of Equation (23) evaluated at whatever θ the sampler is currently trying.

The injected ("true") PSD does not appear in the likelihood at all. It was only used to generate the data in Section 6. When $\theta = \theta_{\text{inj}}$, the model coincides with it.

8.2 Comparison with the textbook Gamma likelihood

The Gamma density with shape $k > 0$ and scale $\vartheta > 0$ is (NIST/SEMATECH e-Handbook, Sec. 1.3.6.6.11; Casella & Berger, *Statistical Inference*, Sec. 3.3)

$$p(x \mid k, \vartheta) = \frac{x^{k-1}\, e^{-x/\vartheta}}{\Gamma(k)\,\vartheta^k}, \qquad \langle x\rangle = k\vartheta, \qquad \mathrm{Var}(x) = k\vartheta^2. \tag{32}$$

For n independent samples with the *same* k and ϑ, the log-likelihood is

$$\ell(k,\vartheta) = (k-1)\sum_{i=1}^{n}\log x_i - \frac{1}{\vartheta}\sum_{i=1}^{n} x_i - n\log\Gamma(k) - nk\log\vartheta. \tag{33}$$

In Equation (31), every pair (j,m) is a *single* observation with its own shape and scale. The sum over channels and bins plays the role of the sum over samples in Equation (33), with $n = 1$ for each term. Writing one term of Equation (33) with the overall minus sign of Equation (31) pulled out:

Textbook, one sample	Eq. (31), one (j,m)	Identification
x	$\hat{P}^c_{jm}$	the data
$(k-1)\log x$	$+\,(\frac{n_{b_m}}{2}-1)\log \hat{P}$	shape $k = n_{b_m}/2$
$-\log\Gamma(k)$	$-\log\Gamma(\frac{n_{b_m}}{2})$	same k
$-x/\vartheta$	$-\frac{n_{b_m}}{2}\,\hat{P}/S$	scale $\vartheta = 2S/n_{b_m}$
$-k\log\vartheta$	$-\frac{n_{b_m}}{2}\log S$	$= -k\log\vartheta$ up to $-\frac{n_{b_m}}{2}\log\frac{2}{n_{b_m}}$

The last entry differs only by a term that depends on n_{b_m} alone. That term does not depend on θ, so Buscicchio, Pozzoli et al. (2025) drops it. In summary,

$$\hat{P}^c_{jm} \sim \text{Gamma}\left(k = \frac{n_{b_m}}{2},\quad \vartheta = \frac{2S^c_{jm}(\theta)}{n_{b_m}}\right), \qquad \langle \hat{P}\rangle = k\vartheta = S^c_{jm}(\theta). \tag{34}$$

Interpretation.

- *The model PSD is the mean, and it sets the scale.* S is not the scale parameter itself but the mean $k\vartheta$. Since $\vartheta = 2S/n_{b_m}$ is proportional to S, changing S at fixed n_{b_m} simply stretches the distribution of $\hat{P}$ without changing its shape.
- *The number of averaged bins sets the shape.* The relative scatter is $\sqrt{\text{Var}}/\langle\hat{P}\rangle = 1/\sqrt{k}$. For small n_{b_m} the distribution is skewed (an exponential for $k = 1$). For large n_{b_m} it approaches a narrow Gaussian centred on S. High-frequency intervals, which average many bins, therefore constrain the model more tightly.
- *What the sampler sees.* Only two terms depend on θ:

$$\log \mathcal{L}^{\text{chunk}}_G = -\sum_{j,m}\frac{n_{b_m}}{2}\left[\log S^c_{jm}(\theta) + \frac{\hat{P}^c_{jm}}{S^c_{jm}(\theta)}\right] + \text{const.} \tag{35}$$

  Each bracket is minimised at $S = \hat{P}$, and each is weighted by n_{b_m}. For $n_{b_m} = 1$ this reduces to the θ-dependent part of the Whittle likelihood, Equation (4). So the Gamma likelihood is the natural generalisation of Whittle to averaged periodograms.

Remark: counting degrees of freedom

A complex Fourier amplitude carries *two* real Gaussian degrees of freedom, so each $|\tilde{d}_{jk}|^2$ is Gamma with shape 1 (as in Buscicchio, Klein et al. (2025), Eq. (21)). The mean of n_{b_m} of them is therefore Gamma with shape n_{b_m} and scale S/n_{b_m}, with variance S^2/n_{b_m}. Equation (31) instead uses shape $n_{b_m}/2$, which corresponds to variance $2S^2/n_{b_m}$. The θ-dependent part of Equation (31) is then exactly one half of the exact one:

$$\mathcal{L}_{\text{Eq. (9)}}(\theta) \propto \mathcal{L}_{\text{exact}}(\theta)^{1/2}.$$

This is equivalent to tempering the likelihood with inverse temperature 1/2. The location of the posterior is unchanged, but its widths are inflated by about $\sqrt{2}$. The Whittle likelihood of Equation (4) carries the same factor $\frac{1}{2}$ per complex amplitude, so the two likelihoods remain consistent with each other. Doubling n_{b_m} in Equation (31), and dropping the $\frac{1}{2}$ in Equation (4), gives the exact versions. We flag this as a point to discuss with the authors.

9 Inference

Putting Equation (1) and Equation (31) together gives $\log \mathcal{L}(\hat{\boldsymbol{P}} \mid \theta)$. Bayes' theorem then gives the posterior $p(\theta \mid \hat{\boldsymbol{P}}) \propto \mathcal{L}(\hat{\boldsymbol{P}} \mid \theta)\, \pi(\theta)$. In Buscicchio, Pozzoli et al. (2025) the parameter vector has 14 components: 4 for the GF spectrum ($\theta_{\mathrm{GF_I}}$), 5 for the envelope ($\theta_{\mathrm{GF_{II}}}$), 3 for the extragalactic foreground (θ_{EF}), and the two noise amplitudes A_{n} and P. All have uniform priors. Without the extragalactic foreground there are 11.

At every step the sampler proposes a trial θ, and only the model branch of Figure 1 is recomputed: $\overline{M}^c_j(\theta)$ for every chunk, the spectra at the coarse frequencies, and the model PSD $S^c_{jm}(\theta)$. The data $\hat{P}^c_{jm}$ never change. Because every ingredient is analytic (the envelope of Equation (20), its chunk average and the spectra), the whole chain from θ to $\log \mathcal{L}$ is cheap and differentiable. Two kinds of sampler are used:

- **Hamiltonian Monte Carlo** with the No-U-Turn Sampler (NUTS, Hoffman & Gelman 2014), as implemented in NumPyro. HMC moves through parameter space along trajectories guided by the gradient $\nabla_\theta \log p(\theta \mid \hat{\boldsymbol{P}})$, which is obtained by automatic (reverse-mode) differentiation in `jax`. No gradient has to be derived by hand. This makes HMC efficient even with many correlated parameters.
- **Nested sampling** with `nessai`, which uses normalising flows to propose new live points. Nested sampling is not an MCMC method. It explores the prior volume shell by shell and returns the Bayesian evidence $\mathcal{Z} = \int \mathcal{L}\, \pi \, \mathrm{d}\theta$ together with the posterior. The evidence is what allows the modulated (quasi-stationary) model to be compared with a stationary one, in which $\overline{M}^c_j$ is constant, via their Bayes factor.

Buscicchio, Pozzoli et al. (2025) finds the same posteriors with both samplers and with both likelihoods (Whittle and Gamma).

10 Summary of the chunked pipeline

The chunked pipeline in nine steps

1. Split the observation into N_c chunks $[t^c_S, t^c_E]$ of duration T; each has the frequency grid $f_k = k/T$.
2. Evaluate the stationary spectra $S^{\mathrm{GF}}(f_k)$ and $S^{(n)}_j(f_k)$ (Eqs. 5–7); they are the same for every chunk.
3. For each chunk and channel, compute the chunk-averaged modulation $\overline{M}^c_j$ (Equation (21)) analytically from the sky-integrated envelope (Equation (20)).
4. Form the total PSD $S^c_{jk} = \overline{M}^c_j S^{\mathrm{GF}}_k + S^{(n)}_{jk}$ (Equation (23)).
5. Draw $\Re \tilde{d}^c_{jk}$ and $\Im \tilde{d}^c_{jk}$ independently from $\mathcal{N}(0, S^c_{jk}/2)$ with the injected parameters (Equation (24)).
6. Group the fine frequencies into n_g log-spaced intervals, average $|\tilde{d}^c_{jk}|^2$ in each, and keep $\hat{P}^c_{jm}$ and n_{b_m} (Equation (30)).
7. For a trial θ, recompute $\overline{M}^c_j(\theta)$ and $S^c_{jm}(\theta)$ at the coarse frequencies.
8. Evaluate the Gamma likelihood (Equation (31)) and sum it over chunks (Equation (1)).
9. Let the sampler (NUTS or nessai) propose the next θ and return to step 7.

Steps 1–6 are done once. Steps 7–9 are repeated until the posterior is sampled.

Part II
The wavelet-domain (WDM) analysis

Notation in Part II. We keep $j \in \{A, E\}$ for the TDI channel and c for a chunk of Part I. A WDM pixel is labelled (n, m), with n its time index and m its frequency index, as in Cornish (2020). (Inside Part II, m is therefore a WDM frequency row, not a coarse bin of Section 7.) Groups of pixels are labelled by a time block s and a frequency group g. The sampling cadence is $\mathrm{d}t$, and N is the total number of samples.

11 Why a wavelet-domain analysis

The chunked analysis of Part I has to answer two questions with one number, the chunk length T. It must be *long*, so that each chunk has fine frequency resolution $1/T$, and *short*, so that the modulation is nearly constant inside it. The modulation then enters only as one averaged value $\overline{M}_j^c$ per chunk, and anything that happens on time scales shorter than a chunk is averaged away.

A wavelet (time–frequency) representation removes this compromise. It describes the data by coefficients that are *local in both time and frequency*, so features that are local in time (a change of the modulation, a glitch, a gap) stay local in the representation instead of being spread over a whole chunk. Cornish (2020) lists the properties that make the WDM transform attractive for gravitational-wave data:

- detector noise is *non-stationary*, which breaks the basic assumption of frequency-domain (Whittle) analyses, whereas for *locally* stationary noise the WDM pixels are, to a good approximation, uncorrelated, which keeps the likelihood diagonal and cheap;
- the variance of each pixel follows the local, time-dependent spectrum $S(f, t)$, so a time-varying spectrum (like the modulated GF) is described pixel by pixel instead of chunk by chunk;
- signals that are compact in time–frequency (binaries, bursts) occupy a small set of pixels, and their wavelet coefficients can be computed directly.

The WDM transform was used for the time-varying Galactic foreground by Digman & Cornish (2022). For data gaps, the point that motivates this work, the relevant property is locality: a gap damages only the pixels near it (Section 20). Gaps still cause leakage at their edges, and Pearson & Cornish (2025) develop Bayesian gap filling in the time–frequency domain to deal with it.

The broader aim here is therefore: *can the Galactic foreground be analysed in the WDM domain as well as with chunks, and does this domain give a natural way of handling data gaps?* This part describes the method; Part III adds the treatment of gaps, and the injection runs are presented in Part V.

Two different "wavelets"

The word is used for two unrelated things, kept apart throughout:

- the **WDM basis functions** $g_{nm}(t)$ (Section 14). They transform the *data*. They are fixed by the grid and have no free parameters.
- the **modulation atoms** ψ (Section 13). They describe the *model* of $P_j(t)$. Their number and their parameters are sampled.

12 The data: one continuous time series

A wavelet analysis needs one continuous time series, not independent chunks. For each TDI channel the simulation draws two independent *stationary* Gaussian time series and modulates only the Galactic one:

$$x_j(t) = n_j(t) + \sqrt{P_j(t)}\, g_j(t). \tag{36}$$

Here n_j has the instrument-noise PSD $S_j^{(n)}(f)$, g_j has the Galactic PSD $S^{\mathrm{GF}}(f)$ of Equation (5), and $P_j(t)$ is the instantaneous power modulation. The order is: draw the stationary series first, then multiply the Galactic one by $\sqrt{P_j(t)}$. Each stationary series is drawn in the frequency domain, with $\langle |X_k|^2 \rangle = NS(f_k)/(2\,\mathrm{d}t)$, and inverse-Fourier transformed. Because P_j changes over days to weeks, it is evaluated hourly and interpolated to the 10-s samples. The series is simulated with two extra days at each end and then cut, so that its ends do not wrap around (a series drawn in the frequency domain is periodic, real data are not).

Which modulation? Two choices are used:

- the **true sky envelope**, $P_j(t) = M_j^2(t)$ of Equation (20). This is the *instantaneous* envelope, a smooth function of time. It is *not* the chunk average $\overline{M}_j^c$ of Equation (21): no chunks are involved anywhere in Part II. (Run 2.)
- a modulation *built from wavelet atoms*, so that the model can represent it exactly (Section 13). (Run 1.)

For a modulation that is slow compared with the WDM time resolution, Equation (36) has the time-dependent ("evolutionary") spectrum

$$S_j(f,t) = P_j(t)\, S^{\mathrm{GF}}(f) + S_j^{(n)}(f). \tag{37}$$

Averaged over a chunk, it reproduces the chunk PSD of Equation (23).

As in Part I, the data are generated once, transformed once and then frozen. During sampling only the model is recomputed.

13 Modelling the modulation with wavelet atoms

13.1 The atom model

The sky model of Section 4 ties $P_j(t)$ to five parameters of an assumed Galactic shape. To measure the modulation *without* assuming that shape, we write its logarithm as a constant plus a sum of K_j localised bumps, the *atoms*:

$$\ln P_j(t) = b_j + \sum_{k=1}^{K_j} a_{j,k}\, \psi\left(\frac{t - t_{j,k}}{\tau_{j,k}}\right), \qquad \psi(x) = e^{-x^2/2}. \tag{38}$$

Each atom has three parameters: a centre $t_{j,k}$ (sampled as the fraction $u_{j,k} = (t_{j,k} - t_0)/T_{\mathrm{obs}}$ of the observation), a width $\tau_{j,k}$ (sampled as $\log_{10}$ of the width in days) and an amplitude $a_{j,k}$. The number of atoms K_j is itself a parameter, and each channel has its own atoms.

Why this choice.

- *Logarithm.* Writing the model for $\ln P$ guarantees $P_j(t) > 0$ for any parameters, and makes each atom a multiplicative change: an atom of amplitude a multiplies the Galactic power at its centre by e^a.

- *Gaussian atoms.* They are smooth, localised (a few τ wide) and described by a centre, a width and a height, which are easy to interpret and to put priors on. A Mexican-hat shape, $\psi(x) = (1 - x^2)e^{-x^2/2}$, is also implemented.
- *Local.* An atom changes the model only near its centre, so a local feature of the modulation can be fitted without changing it elsewhere. This is what a global sky model cannot do.
- *Variable number.* How many atoms the data need is not known in advance. Treating K_j as a parameter lets the data decide (Section 19).

What is measured. The Galaxy enters the data only through $S^{\mathrm{GF}}(f)\,P_j(t) \propto A\,P_j(t)$, with $A = 10^{\mathrm{amp}}$ the Galactic amplitude. The constant b_A is therefore degenerate with the amplitude and is fixed to 0; b_E stays free and measures the E level relative to A. Even then, an atom much wider than the data is almost a constant that the amplitude can absorb. Only the product $A\,P_j(t)$ is identified, and that is the quantity compared with the truth.

13.2 Run 1: can the atoms be recovered?

Before analysing the true envelope we test the method on data whose modulation the atom model represents *exactly*:

1. **Fit once.** Fit Equation (38) by least squares to the true envelope $\ln M_j^2(t)$ over the 30 days, for $K = 0, 1, 2, \ldots$, and keep the smallest K with an rms error below 1% in P. A small penalty on the amplitudes picks one solution along the width/level degeneracy, and the centres are kept away from the ends of the data so that no injected value sits on a prior boundary. (In the runs so far: two atoms in A, one in E.)
2. **Inject.** Use the fitted atoms as $P_j(t)$ in Equation (36) and simulate the data.
3. **Recover.** Run the inference of Section 19, starting with no atoms, and check whether it finds the number of atoms, the atoms and, above all, $A\,P_j(t)$.

The fit happens only once, to build the injection. During inference nothing is fitted: a set of atoms proposed by the sampler *defines* $P_j(t)$ through Equation (38), and the likelihood says how well that $P_j(t)$ explains the frozen data. Run 2 then injects the true envelope $M_j^2(t)$, which no finite set of atoms represents exactly, and analyses the same data with the chunked model of Part I for comparison.

14 From the STFT to the WDM transform

14.1 The time–frequency plane and its pixels

A plain Fourier transform of the whole series tells us which frequencies are present, but not when: each Fourier coefficient is spread over the whole observation. A time–frequency representation cuts the plane (time horizontally, frequency vertically) into boxes and gives, for each box, a number that measures how much signal there is at that time and frequency.

The boxes cannot be arbitrarily small. A stretch of data of duration T resolves frequencies spaced by $1/T$, so making a box shorter in time makes it taller in frequency: only its *area* is fixed. Equivalently (Nyquist–Shannon), a stretch of duration T and bandwidth B holds about $2TB$ independent real numbers. The smallest box that holds exactly *one* real number therefore has area

$$\Delta T\,\Delta F = \tfrac{1}{2}. \tag{39}$$

Such a box is a **pixel**. As in a digital image, a pixel is a *location* (a box in the plane) together with *one value*. Nothing physical is "inside" the box: it labels the region of the plane that the number describes.

14.2 The short-time Fourier transform and its dilemma

The chunked analysis of Part I is a short-time Fourier transform (STFT): cut the series into segments, Fourier transform each one, and obtain a grid of (segment) × (frequency). Two variants show the problem.

- **Rectangular window, no overlap** (the chunks of Part I). The coefficients are independent and there are exactly as many real numbers as samples. But the hard cut at the edges of each segment *leaks* power from one frequency into many others. On a realistic, non-periodic 30-day series, 15-day rectangular chunks overestimate the PSD by 36–72% at 0.13 mHz, until a taper is applied.
- **Smooth window** (Hann, Meyer, ...). Leakage is cured, but the window goes to zero at the segment edges, so the data there get no weight. To use all the data, the segments must *overlap*. Overlapping segments see the same samples: their coefficients are *correlated*, and there are more numbers than samples (the representation is *redundant*). Treating them as independent in a likelihood would count the same data more than once.

A transform is **critically sampled** when it turns N real samples into exactly N real coefficients. With fewer, information is lost; with more, the coefficients are redundant and correlated; with exactly N and an orthonormal basis, every coefficient is independent and the transform is exactly invertible.

The **Balian–Low theorem** (see Daubechies, *Ten Lectures on Wavelets*) says that this cannot be had with complex exponentials: a critically sampled STFT basis of windowed $e^{2\pi\imath ft}$ cannot be orthonormal *and* have a window that is well localised in both time and frequency. A rectangular window keeps orthonormality and loses localisation; a smooth window keeps localisation and loses orthonormality at critical sampling. *This is the problem that putting a Meyer window into an STFT does not solve.*

14.3 Wilson's trick: alternating cosines and sines

The WDM basis avoids the theorem by changing two things (Daubechies, Jaffard & Journé 1991; Necula et al. 2012):

1. **Real waves instead of complex ones.** A complex STFT coefficient is two real numbers (a cosine part and a sine part). WDM keeps only *one* of them at each grid point, a cosine or a sine, alternately. To compensate, the frequencies are spaced *twice as densely*, $\Delta F = 1/(2\Delta T)$ instead of $1/\Delta T$. The count of real numbers stays at N.
2. **A smooth window whose overlaps cancel.** The basis functions are wave packets

$$g_{nm}(t) = \sqrt{2}\,\phi(t - n\Delta T) \times \begin{cases} \cos(2\pi f_m t), & n+m \text{ even}, \\ \sin(2\pi f_m t), & n+m \text{ odd}, \end{cases} \qquad f_m = m\,\Delta F, \tag{40}$$

   (up to sign conventions), with ϕ the smooth Meyer window of Cornish (2020), Eq. (11). Its Fourier transform is flat for $|f| < \Delta F/4$ and falls smoothly to zero at $|f| = \frac{3}{4}\Delta F$; in time it decays over a few ΔT. So each packet *overlaps* its neighbours, both in time and in frequency.

Orthogonality survives the overlaps because of symmetry. Take $t_n = n\Delta T$. Since $f_m t_n = mn\,\Delta F \Delta T = mn/2$,

$$\cos\big(2\pi f_m(t_n+\tau)\big) = (-1)^{mn}\cos(2\pi f_m\tau), \qquad \sin\big(2\pi f_m(t_n+\tau)\big) = (-1)^{mn}\sin(2\pi f_m\tau),$$

so around its own centre a cosine packet is *even* and a sine packet is *odd* (ϕ is even). Then:

- *Neighbours in frequency* (n; m and $m+1$): $n+m$ and $n+m+1$ have opposite parity, so one packet is even and the other odd about the same centre. Their product is odd and integrates to exactly zero.
- *Two rows apart* (m and $m+2$): the frequency supports $|f - f_m| < \frac{3}{4}\Delta F$ are $2\Delta F$ apart and do not overlap at all.
- *Neighbours in time* (n and $n+1$; same m): one is a cosine, the other a sine, and $\cos \cdot \sin = \frac{1}{2}\sin(4\pi f_m t)$. The product of the two windows contains only frequencies below $\frac{3}{2}\Delta F$, so it has no component at $2f_m \geq 2\Delta F$ and the integral vanishes.
- *Diagonal neighbours* and *packets two steps apart in time*: the same arguments around the midpoint between the packets, together with the shape of the Meyer window, which is designed for these cancellations (proof in Necula et al. and Daubechies et al.).

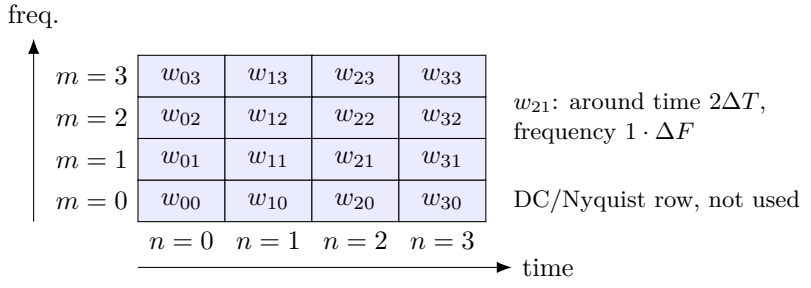
In a complex STFT every packet contains both a cosine and a sine part, and these symmetry arguments have nothing to act on. We checked the result numerically: building every basis function of a small grid, $\sum_l g_{nm}(t_l) g_{n'm'}(t_l) = \delta_{nn'}\delta_{mm'}$ to 10^{-15}.

14.4 The grid: N_t, N_f and the indices n, m

A series of N samples at cadence $\mathrm{d}t$ is mapped to a grid of N_t time columns and N_f frequency rows, with $N_t N_f = N$:

$$\Delta T = N_f\,\mathrm{d}t, \qquad \Delta F = \frac{1}{2N_f\,\mathrm{d}t}, \qquad \Delta T\,\Delta F = \frac{1}{2}, \qquad t_n = n\,\Delta T, \quad f_m = m\,\Delta F. \tag{41}$$

$n = 0, \ldots, N_t - 1$ is the column (time) index and $m = 0, \ldots, N_f - 1$ the row (frequency) index; they are simply the coordinates of a pixel. N_f is the one knob we choose: a larger N_f gives longer pixels with finer frequency resolution. Row $m = 0$ is special (it holds the zero-frequency and Nyquist components) and is never used. For example, $N = 16$ samples with $N_f = 4$ give a 4×4 grid:



In the injection runs, $T_{\mathrm{obs}} = 30\,\mathrm{d}$ and $\mathrm{d}t = 10\,\mathrm{s}$ give $N = 259{,}200$, and $N_f = 3600$ gives $N_t = 72$ pixels of $\Delta T = 10\,\mathrm{h}$ and $\Delta F = 13.9\,\mu\mathrm{Hz}$. The same data in the two tilings:

	STFT, two 15-day chunks (Part I)	WDM, $N_f = 3600$
time width of a box	15 d	10 h
frequency height of a box	0.77 μHz	13.9 μHz
values	$2 \times 64{,}800$ complex	72×3600 real
real numbers in total	$\approx 2 \times 64{,}800 \times 2 = 259{,}200$	259,200

Both carry exactly the information of the 259,200 samples. The STFT spends it on frequency resolution; WDM trades frequency resolution for a time resolution 36 times better, which is what a modulation that changes over days needs. The count also explains a factor that will matter in Section 18: a complex STFT value is *two* real numbers, a WDM pixel is *one*.

14.5 Computing the coefficients

Definition. Each coefficient is one inner product with the *whole* series:

$$w_{j,nm} = \sum_{l=0}^{N-1} x_j(t_l)\, g_{nm}(t_l). \tag{42}$$

It is one *real* number that measures how much the data look like the wave packet g_{nm}, i.e. how much they oscillate at frequency f_m around time t_n. Nothing is cut into segments: the time localisation comes from the shape of g_{nm}, which is negligible far from t_n. Evaluated literally, Equation (42) costs about N^2 operations.

The fast algorithm (Cornish (2020), Eq. (13)). `bahamas` computes the same numbers in the frequency domain, where the window is defined. The STFT works *time first* (cut a segment, then Fourier transform it); this algorithm works *frequency first*:

1. Fourier transform the *whole* series *once*: $X(f)$, on bins spaced $1/(N\mathrm{d}t)$.
2. For row m, keep the Fourier bins around f_m and multiply them by the window $\tilde{\phi}(f - f_m)$. This is a band-pass filter of width about ΔF. One row of pixels is $\Delta F/(1/N\mathrm{d}t) = N_t/2$ Fourier bins tall.
3. Shift the band to zero frequency and inverse Fourier transform it, with length N_t. This gives the band's slowly varying signal at the N_t pixel times $t_0, \ldots, t_{N_t-1}$.
4. From each of these N_t complex values keep *one* real number: the real part if $n + m$ is even (the cosine packet), the imaginary part if $n + m$ is odd (the sine packet). This is the alternation of Equation (40), and it keeps the total count at N.
5. Repeat for every row m.

One inverse FFT gives a whole row, not one pixel. The time index n is the output index of the inverse FFT. Time is not lost in the spectrum: it is stored in the *phases* of the Fourier coefficients (a feature at t_0 contributes a phase $e^{-2\pi\imath f t_0}$ that rotates from bin to bin). Inverse-transforming all N bins recovers N samples at spacing $\mathrm{d}t$; inverse-transforming the N_t bins of one band recovers that band at the coarser spacing $T_{\mathrm{obs}}/N_t = \Delta T$, which is enough because a narrow band changes slowly. Formally, by Parseval's theorem Equation (42) equals

$$w_{nm} \;\propto \sum_{f \text{ near } f_m} X(f)\, \tilde{\phi}(f - f_m)\, e^{+2\pi\imath (f - f_m)\, t_n},$$

and the factor $e^{2\pi\imath(f-f_m)t_n}$, which moves the packet to time t_n, is exactly the kernel of an inverse FFT with output index n. So the window picks the band (m) and the inverse FFT slides the packet along time (n). The transform is therefore a *filter bank*: N_f narrow bands, each sampled at N_t times. In the injection runs, a row is 36 Fourier bins tall, the code takes 71 bins around f_m, and each inverse FFT has length 72.

14.6 Properties of the WDM transform

- **Orthonormal and critically sampled:** N real samples in, N real coefficients out, $\sum w^2 = \sum x^2$, exactly invertible.
- **Real coefficients**, one real Gaussian degree of freedom per pixel for Gaussian data.
- **Uniform, adjustable tiling:** all pixels have the same ΔT and ΔF, and N_f trades one against the other.

- **Localised:** each basis function is compact in frequency ($|f - f_m| < \frac{3}{4}\Delta F$) and decays quickly in time (a few ΔT), so there is little leakage.
- **Nearly diagonal for locally stationary noise:** pixels are almost uncorrelated and their variance follows $S(f, t)$ (Cornish (2020); Section 15).
- **Fast:** one FFT of length N and N_f inverse FFTs of length N_t.
- **Periodic edges:** the FFT treats the series as periodic, so the first and last time columns mix the end of the series with its start and are discarded (Section 15).

Our implementation follows Cornish (2020) and agrees with the `pywavelet` package coefficient by coefficient, up to a global factor $\sqrt{2}$ in their normalisation (theirs is not orthonormal).

15 Statistics of a WDM pixel

The likelihood rests on one statement:

$$\boxed{w_{j,nm} \sim \mathcal{N}(0, \sigma^2_{j,nm}), \qquad \sigma^2_{j,nm} = \frac{P_j(t_n)\,\langle S^{\mathrm{GF}}\rangle_m + \langle S_j^{(n)}\rangle_m}{2\,\mathrm{d}t},} \tag{43}$$

with different pixels independent. We justify this in four steps.

1. Gaussian, with zero mean. Equation (42) is a weighted sum of data samples. The samples are Gaussian (both processes in Equation (36) are Gaussian, and multiplying by the fixed curve $\sqrt{P_j(t)}$ keeps them Gaussian), and a weighted sum of Gaussians is Gaussian. Every term has zero mean. This is the same reason the Fourier amplitudes of Part I are Gaussian: the FFT is also a weighted sum.

2. Variance $S/(2\mathrm{d}t)$. For white noise of variance σ_x^2 per sample, the one-sided PSD S is flat up to the Nyquist frequency $1/(2\mathrm{d}t)$ and integrates to the variance, $\sigma_x^2 = S/(2\mathrm{d}t)$. An orthonormal transform is a rotation in N dimensions and maps independent $\mathcal{N}(0, \sigma_x^2)$ samples to independent $\mathcal{N}(0, \sigma_x^2)$ coefficients. So $\mathrm{Var}(w) = S/(2\mathrm{d}t)$. The same follows from counting energy: a band of width ΔF carries power $S\,\Delta F$; over $\Delta T/\mathrm{d}t$ samples that is an energy $S\Delta F\Delta T/\mathrm{d}t = S/(2\mathrm{d}t)$, and the orthonormal transform puts it into the one coefficient of the box. For coloured noise a pixel sees the PSD through its window, so $S(f_m)$ is replaced by the window average

$$\langle S\rangle_m = \frac{\int S(f)\,|\tilde{\phi}(f - f_m)|^2\,\mathrm{d}f}{\int |\tilde{\phi}(f)|^2\,\mathrm{d}f}, \tag{44}$$

evaluated in the code on 17 points across $|f - f_m| < \frac{3}{4}\Delta F$. This matters at low frequency, where the spectrum is steep and a pixel ($13.9\,\mu$Hz) is 14% of the frequency at 0.1 mHz. The cosine and sine packets carry the same energy, so the parity of $n + m$ does not affect the variance.

3. The modulation enters as $P_j(t_n)$. The two components of Equation (36) are independent, so their variances add. The Galactic part of w_{nm} is $\sum_l g_{nm}(t_l)\sqrt{P_j(t_l)}\,g_j(t_l)$. The packet is non-zero only within a few ΔT of t_n (about a day), over which P_j hardly changes, so $\sqrt{P_j(t_n)}$ can be taken out of the sum. This is the *locally stationary* approximation; its error grows with the curvature of P_j over a day.

4. Independence (the diagonal approximation). For white stationary noise the pixels are exactly independent (orthonormality). For coloured or modulated noise the cancellations of Section 14 are exact only if the spectrum is flat across the overlap of two packets, so small correlations could remain. We measured them on 100 fresh simulations of the run-1 setup (30 d, $N_f = 3600$; the injected atoms make the expected value exact). Correlated pixels would show up as a variance of the grouped statistic above the Gamma value $2/\nu_g$ (see Section 17):

band	median ν_g	mean $Y/\langle Y\rangle$ (expect 1)	$\nu_g \,\mathrm{Var}(Y/\langle Y\rangle)$ (expect 2)	corr. freq / time (expect 0)
0.1–0.3 mHz	2	1.005	2.010 ± 0.024	$+0.003$ / $+0.001$
0.3–1 mHz	2	1.002	1.997 ± 0.010	-0.002 / -0.003
1–3 mHz	6	1.002	2.007 ± 0.009	$+0.000$ / -0.000
3–10 mHz	22	1.000	1.999 ± 0.006	$+0.000$ / -0.000
10–29 mHz	69	1.001	2.006 ± 0.007	$+0.000$ / $+0.000$

The effective number of degrees of freedom equals ν_g to 1% in every band, even at the lowest frequencies. The same data give w^2/σ^2 with mean 1 and variance 2, as χ^2_1 requires. The check covers data simulated from the model itself (no gaps or glitches) and this grid only.

Edges. Because the transform is periodic, the first and last time columns are biased (by +10% over the band, and up to +270% below 1 mHz, in the first column). From the third column on, every pixel is unbiased, so the likelihood drops two columns (20 h) at each end (`edge_bins`). An end of the series behaves like a gap; Section 20 returns to this.

16 Coarse graining: grouping pixels

The 30-day analysis has $2 \times 68 \times 2081 \approx 283{,}000$ pixels in $[0.1, 29]$ mHz. As in Section 7, they are coarse grained to make each likelihood evaluation cheaper. A **group** is a rectangle of pixels:

$$\text{group } (j,s,g) = \underbrace{\texttt{time_bin} \text{ consecutive time columns}}_{\text{time block } s} \times \underbrace{\text{the rows whose } f_m \text{ falls in log bin } g}_{\text{frequency group } g},$$

and the data enter only through the sum of squares over it,

$$Y_{j,sg} = \sum_{n\in s} \sum_{m\in g} w^2_{j,nm}, \qquad \nu_g = \text{number of pixels in the group.} \tag{45}$$

In the injection runs `time_bin` = 2 (blocks of 20 h) and the band is split into 200 log-spaced bins of equal width in $\log f$ (2.9% each). At low frequency a log bin is narrower than one pixel, so it contains one row or none; empty bins are dropped, leaving 174 groups. ν_g grows from 2 (one row $\times$ two columns) at the bottom to 116 at the top. The likelihood then has $2 \times 34 \times 174 \approx 11{,}800$ terms. Neither 200 nor 2 was tuned; they are a speed–accuracy compromise (see the end of this section).

Two different averages of the model. On the data side, Equation (45) is a plain sum. On the model side, two averages appear and should not be confused:

1. *Inside one pixel*, the window average $\langle S\rangle_m$ of Section 15. It describes what a single pixel measures and is needed even without grouping.
2. *Across the pixels of a group*, the plain mean $\bar{S}_g = \nu_g^{-1} \sum_{m\in g} \langle S\rangle_m$ (counted once per row). It exists only because of grouping.

The modulation is *not* averaged over a block: it is evaluated once, at the block's mean time $\bar{t}_s$. The model variance of a group is

$$\sigma^2_{j,sg} = \frac{P_j(\bar{t}_s)\,\bar{S}^{\mathrm{GF}}_g + \bar{S}^{(n)}_{j,g}}{2\,\mathrm{d}t}. \tag{46}$$

	chunked (Part I)	wavelet
data in one term	$\hat{P}$ = mean of $\lvert\tilde{d}_k\rvert^2$ over a log bin	Y = sum of w^2 over a rectangle
model within one value	S at the bin midpoint	window average $\langle S\rangle_m$ per pixel
model across the term	one midpoint value	mean of $\langle S\rangle_m$ over the group
time dependence	$\overline{M}^c_j$, averaged over a 15-day chunk	$P_j(\bar{t}_s)$, at the centre of a 20-h block

Are the variances inside a group equal? The model assumes so (one σ^2 per group); in reality the PSD changes slightly from row to row and P_j across a block. For the run settings the largest spread of σ^2 inside a group is 0.3% below 0.3 mHz, 3–4% up to 3 mHz, 10% above 10 mHz and 15% around the knee of the Galactic spectrum (3–10 mHz); P_j changes by at most 1.7% over a block. Three consequences:

- the mean is exact: $\langle Y\rangle = \sum \sigma^2_{nm} = \nu_g\bar{\sigma}^2$ whatever the spread;
- the variance of Y exceeds the Gamma value by the factor $1 + \mathrm{CV}^2$, with CV the relative spread of the σ^2: at most 1.002 here;
- information can be lost. If the ν_g variances are truly equal, Y is a *sufficient statistic*: summing loses nothing. If they differ, how σ^2 varies *inside* the group is discarded. The 15% spread around the knee may cost some precision on f_{knee}; a run without grouping would settle it.

17 Why the grouped statistic is Gamma distributed

We now show, step by step, that if $w_1, \ldots, w_\nu$ are independent and each $w_i \sim \mathcal{N}(0, \sigma^2)$ with the *same* σ^2, then

$$Y = \sum_{i=1}^{\nu} w_i^2 \;\sim\; \mathrm{Gamma}\!\left(k = \frac{\nu}{2},\ \vartheta = 2\sigma^2\right). \tag{47}$$

Step 1: the tool. The moment generating function (MGF) of a random variable X is $M_X(t) = \langle e^{tX}\rangle$, a function of the real number t. Two facts are all we need:

(A) *It is a fingerprint:* two distributions with the same MGF (on an interval around $t = 0$) are the same distribution.

(B) *Sums become products:* if X_1 and X_2 are independent,

$$M_{X_1+X_2}(t) = \langle e^{tX_1}e^{tX_2}\rangle = \langle e^{tX_1}\rangle\langle e^{tX_2}\rangle = M_{X_1}(t)\,M_{X_2}(t).$$

The middle equality, "the mean of a product of independent quantities is the product of their means", is the only place where independence is used.

Step 2: one squared pixel. With the Gaussian density of w,

$$\langle e^{tw^2}\rangle = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-w^2/(2\sigma^2)}\, e^{tw^2}\, \mathrm{d}w = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-w^2\left(\frac{1}{2\sigma^2}-t\right)}\, \mathrm{d}w.$$

For $t < 1/(2\sigma^2)$ the integrand is again a Gaussian shape, with a new variance σ'^2 defined by $\frac{1}{2\sigma'^2} = \frac{1}{2\sigma^2} - t$, i.e. $\sigma'^2 = \sigma^2/(1-2\sigma^2 t)$. Using $\int e^{-w^2/(2\sigma'^2)}\mathrm{d}w = \sqrt{2\pi\sigma'^2}$,

$$\langle e^{tw^2}\rangle = \sqrt{\frac{\sigma'^2}{\sigma^2}} = (1-2\sigma^2 t)^{-1/2}.$$

Step 3: the sum. By (B), the MGF of Y is the product of ν identical factors:

$$M_Y(t) = \left[(1-2\sigma^2 t)^{-1/2}\right]^{\nu} = (1-2\sigma^2 t)^{-\nu/2}.$$

This step also needs all ν variances to be *equal*; otherwise the factors differ and do not combine into one power.

Step 4: the MGF of a Gamma distribution. With the density of Part I, $p(x \mid k,\vartheta) = x^{k-1}e^{-x/\vartheta}/(\Gamma(k)\vartheta^k)$ for $x>0$, and the standard integral $\int_0^\infty x^{k-1}e^{-ax}\mathrm{d}x = \Gamma(k)/a^k$,

$$M(t) = \frac{1}{\Gamma(k)\vartheta^k}\int_0^\infty x^{k-1}e^{-x\left(\frac{1}{\vartheta}-t\right)}\mathrm{d}x = \frac{1}{\Gamma(k)\vartheta^k}\,\frac{\Gamma(k)}{(1/\vartheta - t)^k} = (1-\vartheta t)^{-k}.$$

Step 5: match. Steps 3 and 4 are the same function when $k=\nu/2$ and $\vartheta = 2\sigma^2$. By (A), Equation (47) holds. Equivalently, $Y/\sigma^2 \sim \chi^2_\nu$, the chi-squared distribution with ν degrees of freedom.

Step 6: sanity check. The Gamma mean is $k\vartheta = \frac{\nu}{2}\cdot 2\sigma^2 = \nu\sigma^2$, as it should be for ν terms of mean σ^2. The variance is $k\vartheta^2 = 2\nu\sigma^4$, so

$$\mathrm{Var}\left(\frac{Y}{\langle Y\rangle}\right) = \frac{2\nu\sigma^4}{\nu^2\sigma^4} = \frac{2}{\nu}, \qquad \text{i.e.} \qquad \nu\,\mathrm{Var}\left(\frac{Y}{\langle Y\rangle}\right) = 2,$$

which is the quantity measured in Section 15.

Assumptions used

(i) each w is Gaussian with zero mean (Step 2; Section 15, point 1); (ii) the ν pixels are independent (Step 3, fact B; checked in Section 15); (iii) they share one variance (Step 3; true to 0.2% in variance, Section 16).

18 The wavelet likelihood and its equivalence with Buscicchio, Pozzoli et al. (2025), Eq. (9)

18.1 The likelihood

Taking the logarithm of the Gamma density with $k=\nu_g/2$ and $\vartheta = 2\sigma^2$, and summing over channels, blocks and groups (independent by Section 15):

$$\log\mathcal{L}(\boldsymbol{Y}\mid\theta) = \sum_{j,s,g}\left[\left(\frac{\nu_g}{2}-1\right)\log Y_{j,sg} - \frac{Y_{j,sg}}{2\sigma^2_{j,sg}} - \log\Gamma\left(\frac{\nu_g}{2}\right) - \frac{\nu_g}{2}\log\left(2\sigma^2_{j,sg}\right)\right], \tag{48}$$

with $\sigma^2_{j,sg}(\theta)$ from Equation (46). This is exactly what the code evaluates (`model.py` in `bahamas/wavelet`; the terms without θ are computed once, the two with σ^2 at every call). As in Part I, Y is the **data** and $\sigma^2(\theta)$ the **model**.

18.2 Rewriting it in the variables of Buscicchio, Pozzoli et al. (2025), Eq. (9)

To compare with Equation (31) we relabel its coarse bin $m \to g$ and its chunk $c \to s$. For one term (indices dropped), Equation (31) reads

$$\ell_{\text{Eq. 9}} = -\left[\log\Gamma\left(\frac{n}{2}\right) + \frac{n}{2}\log S - \left(\frac{n}{2}-1\right)\log\hat{P} + \frac{n}{2}\frac{\hat{P}}{S}\right], \tag{49}$$

and one term of Equation (48) is

$$\ell_{\text{wdm}} = \left(\frac{\nu}{2}-1\right)\log Y - \frac{Y}{2\sigma^2} - \log\Gamma\left(\frac{\nu}{2}\right) - \frac{\nu}{2}\log(2\sigma^2). \tag{50}$$

We look for a data variable $\hat{P}_w$ (built from Y) and a model variable S_w (built from σ^2) that turn Equation (50) into Equation (49) with $n \to \nu$.

Step 1: the $\log S$ term. Equation (50) contains $-\frac{\nu}{2}\log\sigma^2$ (from $-\frac{\nu}{2}\log 2\sigma^2 = -\frac{\nu}{2}\log 2 - \frac{\nu}{2}\log\sigma^2$), and Equation (49) contains $-\frac{\nu}{2}\log S_w$. They agree up to a constant for *any* proportionality

$$S_w = \kappa\,\sigma^2, \qquad \kappa > 0 \text{ a constant},$$

because $-\frac{\nu}{2}\log S_w = -\frac{\nu}{2}\log\sigma^2 - \frac{\nu}{2}\log\kappa$. This term alone does not fix κ.

Step 2: the data term. Matching $-Y/(2\sigma^2)$ with $-\frac{\nu}{2}\hat{P}_w/S_w$ and inserting $S_w = \kappa\sigma^2$:

$$\frac{Y}{2\sigma^2} = \frac{\nu}{2}\frac{\hat{P}_w}{\kappa\sigma^2} \quad\Longrightarrow\quad \hat{P}_w = \frac{\kappa\,Y}{\nu}.$$

So the whole family $(\hat{P}_w, S_w) = (\kappa Y/\nu,\ \kappa\sigma^2)$ reproduces the two θ-dependent terms of Equation (49).

Step 3: the remaining terms. With $Y = \nu\hat{P}_w/\kappa$,

$$\left(\frac{\nu}{2}-1\right)\log Y = \left(\frac{\nu}{2}-1\right)\log\hat{P}_w + \left(\frac{\nu}{2}-1\right)\log\frac{\nu}{\kappa},$$

and $-\log\Gamma(\nu/2)$ is common to both. Collecting everything,

$$\ell_{\text{wdm}} = \ell_{\text{Eq. 9}}\Big|_{n\to\nu,\ \hat{P}\to\hat{P}_w,\ S\to S_w} + C_\nu, \qquad C_\nu = \left(\frac{\nu}{2}-1\right)\log\frac{\nu}{\kappa} + \frac{\nu}{2}\log\frac{\kappa}{2}. \tag{51}$$

C_ν contains neither θ nor anything that changes during sampling, so it does not affect the posterior.

Step 4: fixing κ by what the symbols mean. In Buscicchio, Pozzoli et al. (2025), Eq. (9), S is the model PSD (in Hz^{-1}) and $\langle\hat{P}\rangle = S$. Using $\langle Y\rangle = \nu\sigma^2$ and Equation (46),

$$\langle\hat{P}_w\rangle = \frac{\kappa}{\nu}\nu\sigma^2 = \kappa\sigma^2 = S_w \quad \text{(always)}, \qquad S_w = \frac{\kappa}{2\,\mathrm{d}t}\left(P_j\bar{S}_g^{\text{GF}} + \bar{S}_{j,g}^{(n)}\right).$$

S_w is the physical PSD only for $\kappa = 2\,\mathrm{d}t$. Hence

$$\boxed{\hat{P}_w = \frac{2\,\mathrm{d}t\,Y}{\nu}, \qquad S_w = 2\,\mathrm{d}t\,\sigma^2 = P_j(\bar{t}_s)\,\bar{S}_g^{\text{GF}} + \bar{S}_{j,g}^{(n)}, \qquad C_\nu = \frac{\nu}{2}\log\frac{\nu}{2} - \log\frac{\nu}{2\,\mathrm{d}t}.} \tag{52}$$

$\hat{P}_w$ is the *wavelet averaged periodogram*: the mean of the ν values $2\mathrm{d}t\,w^2$, each of which has mean $\langle S\rangle_m$, exactly as each $|\tilde{d}_k|^2$ has mean S. The constant has two parts: $\frac{\nu}{2}\log\frac{\nu}{2}$, which Buscicchio, Pozzoli et al. (2025) also drops (Section 8), and $-\log\frac{\nu}{2\mathrm{d}t}$, the Jacobian of the change of variables $Y \to \hat{P}_w$.

A pitfall. Another member of the family, $\kappa = \nu\,\mathrm{d}t$, gives $\hat{P}_w = \mathrm{d}t\,Y$. It reproduces Equation (49) algebraically, but its model is $S_w = \nu \mathrm{d}t\,\sigma^2 = \frac{\nu}{2}(P_j \bar{S}_g^{\mathrm{GF}} + \bar{S}_{j,g}^{(n)})$, which is not the PSD unless $\nu = 2$. Since the low-frequency groups have $\nu = 2$, matching on one of them gives $\mathrm{d}t\,Y$ by coincidence; a group with $\nu = 69$ shows the difference.

Result. Summing over j, s, g,

$$\log \mathcal{L}_{\mathrm{wdm}}(\boldsymbol{Y} \mid \theta) = \sum_s \log \mathcal{L}_G^{\mathrm{chunk}}(\hat{\boldsymbol{P}}_{w,s} \mid \theta)\Big|_{n_b \to \nu} + \text{const}: \tag{53}$$

the wavelet likelihood is Buscicchio, Pozzoli et al. (2025), Eq. (9) applied to wavelet periodograms, with time blocks in the role of chunks. We checked numerically that the code's value equals the right-hand side to 10^{-12}.

Buscicchio, Pozzoli et al. (2025)	wavelet
chunk c (15 d)	time block s (20 h)
coarse bin with n_b Fourier values	frequency group g with ν_g pixels
$\hat{P} = \frac{1}{n_b}\sum \lvert\tilde{d}_k\rvert^2$ (data)	$\hat{P}_w = \frac{2\mathrm{d}t}{\nu_g}\sum w^2$ (data)
$S = \overline{M}_j^c S^{\mathrm{GF}} + S_j^{(n)}$ (model)	$S_w = P_j(\bar{t}_s)\bar{S}_g^{\mathrm{GF}} + \bar{S}_{j,g}^{(n)}$ (model)
shape $n_b/2$	shape $\nu_g/2$

18.3 The degrees of freedom: an error in Buscicchio, Pozzoli et al. (2025), Eq. (9)

The forms match, but the shape must be half the number of *real* Gaussian degrees of freedom averaged, and the two domains count differently.

- **WDM.** Each pixel is one real Gaussian, so $Y/\sigma^2 \sim \chi^2_\nu$ and the shape $\nu_g/2$ is exact (Section 17).
- **Fourier.** Each $\tilde{d}_k$ is complex; in the simulation of Section 6 its real and imaginary parts are independent, each $\mathcal{N}(0, S/2)$. Repeating Steps 2–5 of Section 17 with $\sigma^2 = S/2$ and *two* real terms per frequency:

$$\lvert\tilde{d}_k\rvert^2 = \tfrac{S}{2}\,\chi^2_2, \qquad \sum_{k=1}^{n_b} \lvert\tilde{d}_k\rvert^2 = \tfrac{S}{2}\,\chi^2_{2n_b}, \qquad \hat{P} = \frac{S}{2n_b}\chi^2_{2n_b} \;\sim\; \mathrm{Gamma}\!\left(k = n_b,\ \vartheta = \frac{S}{n_b}\right),$$

  with $\mathrm{Var}(\hat{P}) = S^2/n_b$. The correct shape is n_b; Buscicchio, Pozzoli et al. (2025), Eq. (9) uses $n_b/2$, which implies $\mathrm{Var}(\hat{P}) = 2S^2/n_b$.

The θ-dependent part of either likelihood is $-(\text{shape})\big[\log S + \hat{P}/S\big]$. Halving the shape halves the curvature of the log-likelihood, which leaves the maximum where it is but makes the posterior widths about $\sqrt{2}$ too large (the remark in Section 8). Counted on the same patch of the time–frequency plane, a region with n_b complex Fourier values holds $2n_b$ real numbers and $\nu = 2n_b$ WDM pixels, so

wavelet	shape $\nu/2 = n_b$	exact
Buscicchio, Pozzoli et al. (2025), Eq. (9) as printed	shape $n_b/2$	half the exact value
Buscicchio, Pozzoli et al. (2025), Eq. (9) corrected	shape n_b	exact

The wavelet likelihood and the *corrected* chunked likelihood agree exactly. Measured on simulated chunked data, $n_b\,\mathrm{Var}(\hat{P}/S) = 1.04 \pm 0.04$, which is the value for shape n_b, against 2 for shape $n_b/2$. Correcting the shape in the chunked runs shrinks their widths by 0.70–0.73 $\approx 1/\sqrt{2}$.

In the code. All three chunked samplers of `bahamas` implement the printed shape $n_b/2$. They are in the folder `bahamas/method`: `setting_hmc.py` (line 112), `setting_nessai.py` (`gamma_lik`) and `setting_eryn.py`. Here n_b is the number of Fourier frequencies averaged in a bin (the `count` returned by the log-binning function `average_log_chunks`). The wavelet likelihood is not affected.

A separate bug in the `eryn` version. `setting_eryn.py` (lines 138–143) evaluates

$$\left(\frac{n_b}{2}-1\right)\log\left(|\hat{P}|^2\right)-\frac{n_b}{2}\log S-\log\Gamma\left(\frac{n_b}{2}\right)-\frac{|\hat{P}|^2}{2S} \quad \text{instead of} \quad \left(\frac{n_b}{2}-1\right)\log\hat{P}-\cdots-\frac{n_b}{2}\frac{\hat{P}}{S}.$$

The data passed to it are already the averaged periodogram $\hat{P}$, a power; the code squares it again and drops the factor n_b. Since $\hat{P}\sim 10^{-40}$ for LISA, the data term $\hat{P}^2/S$ is negligible and the log-likelihood reduces to $-\frac{n_b}{2}\sum\log S$, which grows without bound as $S\to 0$. Any chunked `eryn` run with the Gamma likelihood is therefore invalid. The HMC and `nessai` versions are correct apart from the shape. Neither issue touches the wavelet pipeline, which uses `eryn` only as a sampler with its own likelihood. Both are to be discussed with the authors before the library is changed.

19 Inference: reversible-jump MCMC

19.1 The posterior

The parameters are the spectral ones, $\theta_{\rm s}=\{\alpha,\mathrm{amp},f_{\rm knee},f_1,f_2,A_{\rm n},P,b_E\}$, and for each channel the number of atoms K_j and the atoms $\zeta_{j,k}=(u_{j,k},\log_{10}\tau_{j,k},a_{j,k})$. The posterior is

$$p(\theta_{\rm s},\{K_j,\zeta_{j,1..K_j}\}\mid\boldsymbol{Y})\;\propto\;\mathcal{L}(\boldsymbol{Y}\mid\theta_{\rm s},\{\zeta\})\;\pi(\theta_{\rm s})\prod_{j\in\{A,E\}}\Big[\pi(K_j)\prod_{k=1}^{K_j}\pi(\zeta_{j,k})\Big], \tag{54}$$

with uniform priors throughout. Its *dimension changes with* K_j, which an ordinary MCMC cannot handle.

19.2 What reversible jump does

Reversible-jump MCMC (Green 1995) adds moves that change the dimension. Here they are *birth* and *death* moves:

- a **birth** proposes $K\to K+1$ by drawing a new atom from its prior;
- a **death** proposes $K\to K-1$ by removing an atom chosen at random.

Each is accepted with a Metropolis–Hastings probability built so that the chain satisfies detailed balance *across* dimensions. Because the new atom is drawn from its prior, its prior density cancels its proposal density, and the acceptance probability reduces to the likelihood ratio times the ratio of move-selection probabilities (with corrections at $K_{\min}$ and $K_{\max}$). A birth is accepted only if the extra atom improves the fit enough; a death only if removing an atom costs little. Between jumps, ordinary moves update the parameters at fixed K.

The chain then visits each K in proportion to its posterior probability. With a uniform prior on K, the Bayes factor between two numbers of atoms is simply their posterior odds, read off by counting samples:

$$\ln\mathcal{B}(K,K')=\ln\frac{p(K_j=K\mid\boldsymbol{Y})}{p(K_j=K'\mid\boldsymbol{Y})},\qquad p(K_j=K\mid\boldsymbol{Y})\approx\frac{\#\{\text{samples with }K_j=K\}}{\#\{\text{samples}\}}. \tag{55}$$

No separate evidence computation is needed for each K. The probability of $K=0$ measures the evidence for any modulation at all.

19.3 Implementation and settings

The sampler is `eryn` (Karnesis et al. 2023), an ensemble sampler. An *ensemble* is a set of *walkers*, each a full point in parameter space, that move together; the number of walkers is a setting, not something that is inferred. The parameters are organised in *branches*:

- `psd`: $\theta_{\rm s}$, one leaf, updated with stretch moves (Goodman & Weare 2010);
- `mod_A`, `mod_E`: one leaf per atom, K_j leaves, updated with a Gaussian random walk (step 2% of each prior width) and with birth/death moves per channel.

Parallel tempering runs copies of the ensemble at inverse temperatures $1 = \beta_0 > \beta_1 > \dots$, each sampling $\mathcal{L}^{\beta}\pi$; hot copies change K more easily and swap states with colder ones. Only the cold chain ($\beta = 1$) samples the posterior.

One step of the sampler. For each walker: take the proposed $\theta_{\rm s}$ and atoms; compute the band-averaged spectra $\bar{S}_g^{\rm GF}$, $\bar{S}_{j,g}^{(n)}$; compute $P_j(\bar{t}_s)$ from Equation (38); form $\sigma^2_{j,sg}$ (Equation (46)); evaluate Equation (48) against the frozen Y; accept or reject. Nothing is fitted and no data are redrawn. The likelihood is written in `jax`: compiled, vectorised over walkers and differentiable. The fixed-dimension models (no modulation, or the sky envelope of Part I evaluated at $\bar{t}_s$) can also be sampled with NUTS.

setting	code default	template (1 yr)	injection runs (30 d)
grid: N_f (ΔT)	—	4096 (11.4 h)	3600 (10 h)
`edge_bins` / `time_bin` / `freq_bins`	4 / 1 / per pixel	4 / 4 / 200	2 / 2 / 200
TDI generation	—	2	1
walkers per temperature	32	32	32
temperatures	1	2	4
burn-in / stored steps	—	1000 / 5000	2000 / 6000
K_j range (uniform)	0–8	0–15	0–8
atom width τ (log-uniform)	5–180 d	5–180 d	5–60 d
atom amplitude a (uniform)	$[-3, 3]$	$[-3, 3]$	$[-3, 3]$
level b_E (uniform)	$[-2, 2]$	$[-2, 2]$	$[-3, 3]$
atom centre u (uniform)	$[0, 1]$	$[0, 1]$	$[0, 1]$
start	$\theta_{\rm s}$ near injection, $K_j = 0$	same	same

Starting at $K_j = 0$ tests whether the sampler finds the modulation by itself. Starting $\theta_{\rm s}$ in a small ball around the injection is a weaker test than a start from the prior, and is worth changing in later runs.

20 The likelihood in the time–frequency plane, and data gaps

Figure 2 shows what the likelihood reads, for channel A of the run-1 data at the injected parameters. The left column has every analysed pixel (68×2081), the right column the groups (34×174). The frequency axis is logarithmic, so low-frequency pixels look taller although all have the same ΔF.

Top row: data ÷ model. By Section 18 the likelihood depends on the data only through this ratio. A single pixel follows χ^2_1 and ranges from 0.01 to 10, so on its own it says almost nothing. Groups scatter by about $\sqrt{2/\nu_g}$: at the top ($\nu_g \sim 70$–116) they are within $\sim 15\%$ of 1 and look pale; at the bottom ($\nu_g = 2$) they stay noisy. At the true parameters there is no pattern. A wrong model would show one; a stationary model, for example, would leave a left-to-right colour gradient in the Galactic band.

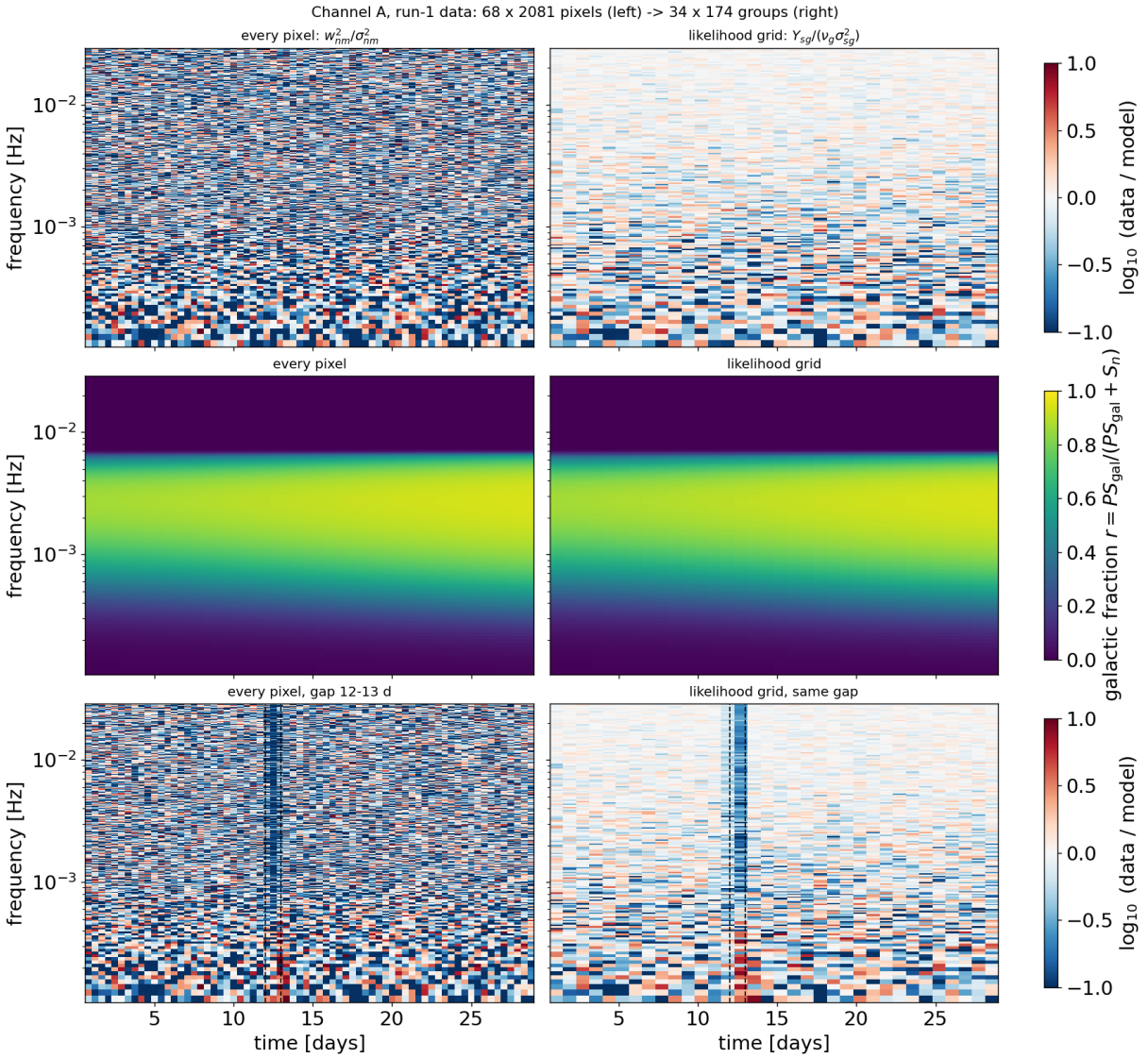


Figure 2: Channel A, run-1 data, at the injection. Top: data/model, w^2/σ^2 per pixel (left) and $Y/(\nu\sigma^2)$ per group (right); white is 1. Middle: Galactic fraction of the variance, $r = P\,S^{\mathrm{GF}}/(P\,S^{\mathrm{GF}} + S^{(n)})$. Bottom: as the top row after setting one day of data (12–13 d, dashed lines) to zero. Produced by `data/wavelet_injection/likelihood_grid.py`.

Middle row: where the Galaxy is learned. The Fisher information on the Galactic amplitude carried by a pixel is $r^2/2$. The Galaxy dominates between about 0.4 and 6 mHz; above the knee (≈ 6.7 mHz) the Galactic model drops steeply and the pixels contain only instrument noise, and below 0.3 mHz the noise dominates again. *The modulation is measured almost entirely in the 0.4–6 mHz band.* Grouping does not move this information; it summarises it.

Bottom row: a one-day gap. The gap was made by setting the samples to zero, with hard edges. Above 1 mHz the pixels inside the gap drop to 0.1–0.4 of the model (the power is missing), and the column on each side is slightly low (0.85–0.94): the damage is *local*, the gap plus about one 10-h pixel on each side, because each packet spans a few ΔT. Below 1 mHz the two columns at the end of the gap are *too high* (about 3 and 2 times the model): zeroing creates a step, a step leaks power over many frequencies, and the leakage stands out where the spectrum is steep. This is the same leakage as with the rectangular chunks of Part I. On the right, a block that only partly overlaps the gap is biased as a whole.

What this suggests for gaps.

1. **Mask, do not fill.** Since the pixels are independent, leaving the affected terms out of the sum is exact. This is what `edge_bins` already does at the two ends of the series. In the Fourier domain a gap breaks the orthogonality of all frequencies of the chunk; in the WDM domain the damage is confined to a strip about a day wide.

2. **Leave a margin**: one pixel on each side above 1 mHz; more below, unless the edges are tapered.

3. **Taper the gap edges** (as the Tukey taper does for the chunks) to remove the low-frequency leakage.

4. **Make the grouping gap-aware**: sum only the unmasked pixels of a group and set ν_g to their number. Equation (47) holds for any ν.

5. **Compared with chunks**: a chunked analysis places its 15-day chunks between gaps, so a short gap can cost most of a chunk; here the unit of data lost is 10 h. The atom model of $P_j(t)$ is continuous across a gap; the side of a gap behaves like an end of the series, where the atom model is least constrained.

Points 1, 2 and 4 are now implemented and are described in Part III; tapering is not. Bayesian gap filling in the time–frequency domain (Pearson & Cornish 2025) is the more complete alternative to masking.

21 Summary of the wavelet pipeline

The wavelet pipeline in eight steps

1. Simulate one continuous series per channel, $x_j = n_j + \sqrt{P_j(t)}\, g_j$ (Equation (36)), with P_j the true envelope or a set of fitted atoms.

2. WDM-transform it once: one FFT, then for each row a windowed band, an inverse FFT of length N_t and the real or imaginary part by the parity of $n+m$ (Section 14.5).

3. Drop the edge columns and the rows outside the band.

4. Group pixels into blocks × log-frequency groups and keep $Y_{j,sg} = \sum w^2$ and ν_g (Equation (45)). Steps 1–4 are done once.

5. For a trial point (spectral parameters, K_j and atoms), compute $P_j(\bar{t}_s)$ from the atoms and the band-averaged spectra.

6. Form $\sigma^2_{j,sg}$ (Equation (46)).

7. Evaluate the Gamma likelihood (Equation (48)), which is Buscicchio, Pozzoli et al. (2025), Eq. (9) with the correct degrees of freedom.

8. Let `eryn` propose the next point, including births and deaths of atoms, and return to step 5.

Status. All injection runs of Part V use the envelope code with the corrections listed in `reports/PAPER_BUGS.md`.

Part III
Data gaps in the WDM domain

22 Injecting a gap and masking it

LISA will not deliver an uninterrupted stream. Antenna repointings, maintenance and unplanned events will interrupt the data for hours to days, and the planned duty cycle is about 75%. Section 20 showed that in the WDM domain a gap damages only the pixels near it. Here we turn that observation into a likelihood and test it on run 2's data with a short gap (run 3) and a long one (run 4).

The gap in the data. We take the 30-day series of run 2 (true sky envelope, same noise realisation) and set it to zero inside the gap:

$$x_j(t_l) \;\to\; 0 \qquad \text{for } t_{\mathrm{gap},1} \le t_l < t_{\mathrm{gap},2}. \tag{56}$$

Run 3 has a 1-day gap, $[12, 13)$ d (8640 samples per channel); run 4 has a 10-day gap, $[10, 20)$ d, a third of the window. Outside the gap the series is identical to run 2's, so any change in the posterior is caused by the gap alone. The zeroed series is then WDM-transformed as before. The gap has hard edges, the worst case for leakage.

Which pixels are damaged. Each WDM packet spans a few ΔT, so pixels whose packets overlap the gap see less than their full share of data. We measured the bias of each pixel's power, $\langle w^2\rangle/\sigma^2$, over 20 noise realisations with a 1-day gap:

- the 3 time columns that overlap the gap are 60–91% low (the power is missing);
- the column just outside on each side is about 10% low above 1 mHz, and 25–40% *high* below 1 mHz, where the step at the gap edge leaks the steep low-frequency spectrum;
- two columns away the bias is below about 2%, the resolution of 20 realisations.

The mask. We define a time mask over the WDM columns,

$$\mu_n = \begin{cases} 0, & \text{column } n \text{ overlaps the gap, or is its first neighbour on either side,} \\ 1, & \text{otherwise,} \end{cases} \tag{57}$$

i.e. the gap columns plus one buffer column (10 h) on each side. The masked columns are *left out* of the likelihood; they are not set to zero. A zero entering Equation (48) would be read as a *measured* power of zero, and would pull the modulation down. The mask is the same idea as `edge_bins`: an end of the series is a gap that never closes.

The masked likelihood. A group of Section 16 is a time block s (two columns) times the r_g frequency rows of log bin g, so $\nu_g = 2r_g$. With the mask, only the kept columns are summed:

$$Y_{j,sg} = \sum_{n\in s} \mu_n \sum_{m\in g} w^2_{j,nm}, \qquad \nu_{sg} = r_g \sum_{n\in s} \mu_n, \qquad \bar{t}_s = \frac{\sum_{n\in s} \mu_n t_n}{\sum_{n\in s} \mu_n}. \tag{58}$$

A block with no kept column is dropped. A block with one kept column keeps half its degrees of freedom, and its modulation is evaluated at the time of that column. The likelihood is Equation (48) with $\nu_g \to \nu_{sg}$ and $P_j(\bar{t}_s)$ at the masked block times. This is *exact*, not an approximation: the derivation of Section 17 needs only that the summed pixels are independent and share one variance, and removing pixels from a group keeps both properties. Equation (47) holds for any ν.

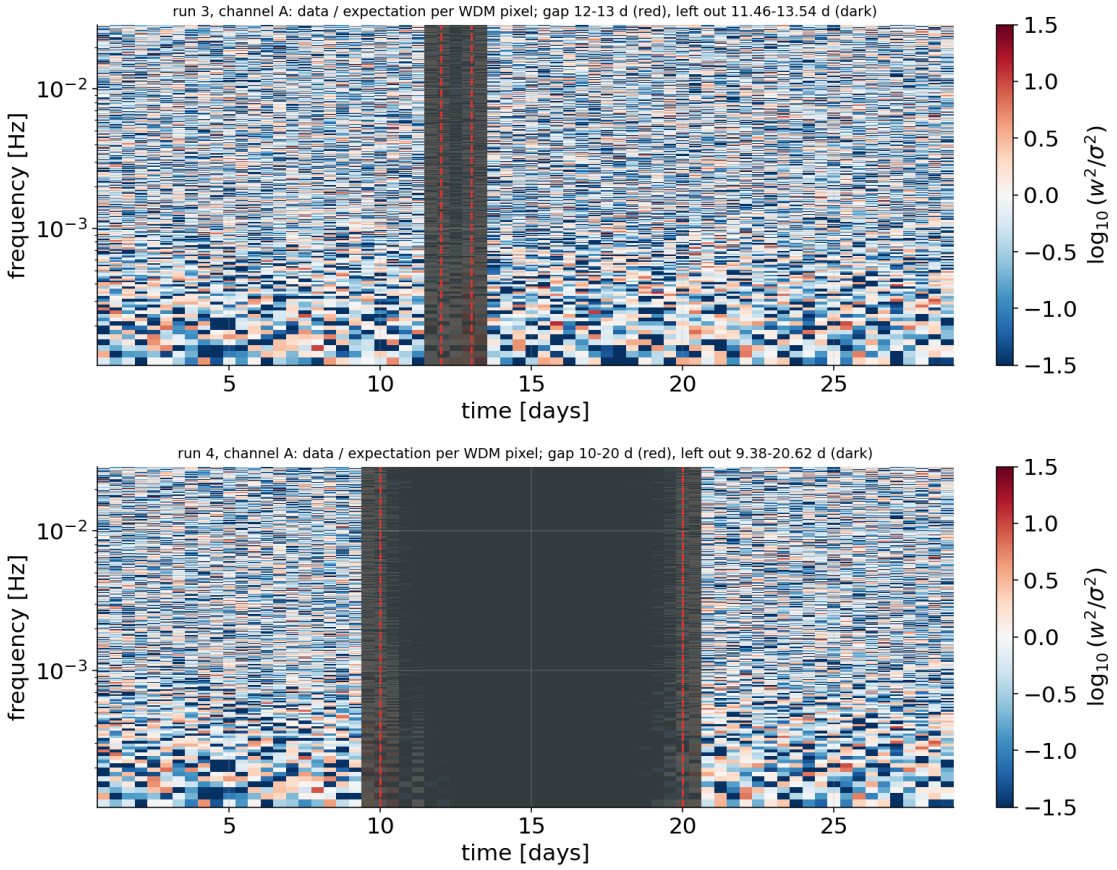


Figure 3: Where the gap is injected. Channel A, data/model per WDM pixel, $\log_{10}(w^2/\sigma^2)$ (white is 1). Red dashed lines: the gap, in which the time series is set to zero. Dark band: the columns left out of the likelihood (gap plus one 10-h buffer column on each side). Top: run 3, 1-day gap. Bottom: run 4, 10-day gap.

	run 3	run 4
gap	12–13 d (24 h)	10–20 d (240 h)
left out (gap + buffer)	11.46–13.54 d (50 h)	9.38–20.62 d (270 h)
time blocks kept	32 of 34	21 of 34
pixels lost	7.4%	39.7%

Figure 3 shows the two set-ups in the time–frequency plane.

23 What happens to the modulation inside the gap

The likelihood has no terms inside the masked region, so the data say nothing about $P_j(t)$ there. The model, however, is defined at every time. Equation (38) writes $\ln P_j(t)$ as a sum of Gaussian atoms, a smooth function of t that exists inside the gap whether or not there are data. Three consequences follow.

1. **Nothing is filled in or interpolated by hand.** The sampler proposes atoms exactly as without a gap; the masked likelihood scores them only on the kept blocks. The atoms are pinned by the data on both sides of the gap, and their values inside the gap follow from their shape.

2. **The prior sets how the gap is bridged.** The atom widths are log-uniform in 5–60 d, so every curve the model can draw is smooth on day scales. Inside a gap the posterior

contains all such curves that join the two sides, and its spread is the honest uncertainty of that interpolation. It is smallest at the gap edges and largest in the middle. A feature of the modulation shorter than the gap could not be recovered inside it; the Galactic envelope has none over 30 days.

3. **The spectral parameters are shared by all times.** They are measured from every kept block, so a gap widens them by roughly $\sqrt{1/(1 - f_{\mathrm{lost}})}$, with f_{lost} the fraction of pixels left out, independently of where the gap is.

When we plot the recovered modulation we show it in two ways: with the gap left blank (only where it is measured), or with the posterior band continued through the gap, as in Figure 10, where its width shows how well the atoms bridge it. The chunked pipeline of Part I has no such choice: chunks are placed between gaps, and the model is evaluated only in them.

Part IV
Needlets and a patchy Galaxy

24 From spherical harmonics to needlets

The atom model of Section 13 describes the modulation in *time*, without reference to the sky. Physically, $P_j(t)$ is the Galaxy's sky distribution seen through LISA's rotating antenna pattern. The Gaussian Galaxy of Section 4 is a crude description of that distribution: it has no disk, no bulge, and none of the satellite galaxies or bar ends that host old stars. To describe the sky without assuming its shape, in the same spirit as the atoms in time, we need a basis of functions on the sphere. This section builds one.

24.1 Spherical harmonics

The spherical harmonics $Y_{\ell m}(\hat{n})$, with multipole $\ell = 0, 1, 2, \ldots$ and $m = -\ell, \ldots, \ell$, are the Fourier modes of the sphere. They are the eigenfunctions of the Laplacian on the sphere, $\nabla^2 Y_{\ell m} = -\ell(\ell+1) Y_{\ell m}$, and are orthonormal and complete:

$$\int Y_{\ell m}\, Y^*_{\ell' m'}\, \mathrm{d}\Omega = \delta_{\ell\ell'}\delta_{mm'}, \qquad f(\hat{n}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{n}), \qquad a_{\ell m} = \int f\, Y^*_{\ell m}\, \mathrm{d}\Omega. \tag{59}$$

The multipole ℓ plays the role of frequency: a harmonic of degree ℓ varies on angular scales of about π/ℓ. Summing over m gives the *addition theorem*,

$$\sum_{m=-\ell}^{\ell} Y_{\ell m}(\hat{n})\, Y^*_{\ell m}(\hat{n}') = \frac{2\ell+1}{4\pi}\, P_\ell(\hat{n}\cdot\hat{n}'), \tag{60}$$

with P_ℓ the Legendre polynomial. So the projection of f onto degree ℓ is $f_\ell(\hat{n}) = \int \frac{2\ell+1}{4\pi} P_\ell(\hat{n}\cdot\hat{n}') f(\hat{n}')\, \mathrm{d}\Omega'$, a kernel that depends only on the angle between the two directions.

Harmonics are perfectly localised in ℓ, but each one covers the whole sky. A compact patch of a few degrees therefore needs many harmonics, and a change in one coefficient changes the map everywhere. The opposite basis, map pixels, is perfectly local on the sky but says nothing about scale. This is the same dilemma as between a Fourier transform and a time series in Section 14, now on the sphere.

24.2 Needlets

Needlets (Narcowich, Petrushev & Ward 2006; Marinucci et al. 2008) are the sphere's analogue of the WDM packets: each is concentrated around one direction *and* in a narrow range of ℓ. They are built in three steps.

1. Split the multipoles into bands. Choose a base $B > 1$ (we use $B = 2$) and a smooth function $\phi(t)$ that equals 1 for $t \le 1/B$, decreases smoothly (C^∞) to 0 at $t = 1$, and is 0 beyond. The window of band $j = 0, 1, 2, \ldots$ is

$$b_j^2(\ell) = \phi\!\left(\frac{\ell}{B^{j+1}}\right) - \phi\!\left(\frac{\ell}{B^{j}}\right), \tag{61}$$

which is non-zero only for $B^{j-1} < \ell < B^{j+1}$. A scaling band $j = -1$ with $b_{-1}^2(\ell) = \phi(\ell)$ collects the largest scales; for $B = 2$ it is the monopole alone. The sum over bands telescopes:

$$b_{-1}^2(\ell) + \sum_{j=0}^{J} b_j^2(\ell) = \phi\!\left(\frac{\ell}{B^{J+1}}\right) = 1 \qquad \text{for } \ell \le B^J. \tag{62}$$

The windows form a smooth partition of unity in ℓ.

2. Build one needlet per band and direction. For each band, take a set of N_j directions ξ_{jk} with weights λ_{jk} (a cubature rule: the pixel centres of a HEALPix grid of resolution $n_{\mathrm{side},j} = 2^j$, with $\lambda_{jk} = 4\pi/N_j$; Górski et al. 2005). The needlet is

$$\boxed{\psi_{jk}(\hat{n}) = \sqrt{\lambda_{jk}} \sum_{\ell} b_j(\ell)\, \frac{2\ell+1}{4\pi}\, P_\ell(\xi_{jk} \cdot \hat{n}).} \tag{63}$$

By Equation (60) it is the band-j part of a delta function at ξ_{jk}, a bump centred on ξ_{jk} and symmetric around it.

3. Analysis and synthesis. The coefficients and the reconstruction are

$$\beta_{jk} = \int f\, \psi_{jk}\, \mathrm{d}\Omega, \qquad f(\hat{n}) = \sum_{j,k} \beta_{jk}\, \psi_{jk}(\hat{n}). \tag{64}$$

The second equation follows in two lines. Inserting Equation (63) twice,

$$\sum_k \beta_{jk}\psi_{jk}(\hat{n}) = \sum_{\ell,\ell'} b_j(\ell) b_j(\ell') \int \mathrm{d}\Omega'\, f(\hat{n}') \sum_k \lambda_{jk} \frac{2\ell+1}{4\pi} P_\ell(\xi_{jk}\cdot\hat{n}') \frac{2\ell'+1}{4\pi} P_{\ell'}(\xi_{jk}\cdot\hat{n}).$$

If the cubature integrates exactly the polynomials that appear (degree up to $2B^{j+1}$), the sum over k is the integral $\int \mathrm{d}\Omega_\xi$, which by Equation (60) and orthogonality gives $\delta_{\ell\ell'} \frac{2\ell+1}{4\pi} P_\ell(\hat{n}'\cdot\hat{n})$. Hence $\sum_k \beta_{jk}\psi_{jk} = \sum_\ell b_j^2(\ell) f_\ell$, and summing over j with Equation (62) returns f.

24.3 Properties

- **Localised in ℓ:** band j contains only $B^{j-1} < \ell < B^{j+1}$ (exactly, by construction).
- **Localised on the sky:** because b_j is smooth, ψ_{jk} decays quasi-exponentially away from ξ_{jk}, over an angle of about π/B^j (Narcowich et al. 2006). Large j means small, sharp needlets.
- **A tight frame, not an orthonormal basis:** Equation (64) reconstructs any band-limited map with the *same* functions for analysis and synthesis, but the needlets overlap and are redundant (there are more coefficients than degrees of freedom). For a prior or a sampler this is harmless; it means a map has more than one set of coefficients.
- **No preferred direction:** every needlet is axisymmetric about its centre, and the construction needs no coordinate patch, unlike wavelets on a flat projection of the sky.
- **Nearly uncorrelated coefficients** for an isotropic random field at high j (Baldi et al. 2009).

With $B = 2$ and our cubature the bands are:

band j	multipoles ℓ	$n_{\mathrm{side},j}$	number of needlets
-1	0	—	1
0	1	1	12
1	2, 3	2	48
2	3–7 ($\ell = 4$ at full weight)	4	192
3	5–15	8	768

This coarse cubature reconstructs a band-limited map to 3.7% (0.6% with twice $n_{\mathrm{side},j}$), enough for a parametrisation.

24.4 Why needlets for the Galactic foreground

LISA sees only $\ell = 0, 2, 4$. In the low-frequency limit the power response of channel j to a source in direction $\hat{n}$, averaged over polarisation, is $R_j(t, \hat{n}) = F^2_{+,j} + F^2_{\times,j}$. Each antenna pattern $F_A = D_{ab}(t)\, e^A_{ab}(\hat{n})$ contracts the detector tensor D_{ab} with a polarisation tensor, and the polarisation sum $\sum_A e^A_{ab} e^A_{cd}$ is built from the projector $\delta_{ab} - \hat{n}_a \hat{n}_b$: a polynomial of degree 4 in the components of $\hat{n}$, even under $\hat{n} \to -\hat{n}$. A polynomial of degree 4 contains only harmonics with $\ell \leq 4$, and an even one only even ℓ. Therefore

$$R_j(t, \hat{n}) = \sum_{\ell \in \{0,2,4\}} \sum_m r_{j,\ell m}(t)\, Y_{\ell m}(\hat{n}), \tag{65}$$

which we confirmed numerically on the first-principles response (power at other ℓ below 10^{-8}). Since $P_j(t) \propto \int I(\hat{n}) R_j(t, \hat{n})\, \mathrm{d}\Omega$, the data see only the projection of the sky onto these $1{+}5{+}9 = 15$ real harmonics. Everything else in the map, including the sharp edges of any patch, is invisible to an A/E auto-power analysis.

Only three bands matter. Expanding the sky in needlets makes $P_j(t)$ linear in the coefficients,

$$P_j(t) = \sum_{j',k} \beta_{j'k}\, T_{j,j'k}(t), \qquad T_{j,j'k}(t) = \frac{\kappa}{\bar{R}} \int \psi_{j'k}(\hat{n})\, R_j(t, \hat{n})\, \mathrm{d}\Omega, \tag{66}$$

(κ and $\bar{R}$ are fixed normalisations, Section 25). T vanishes for every band without $\ell = 0, 2, 4$ content: $j = 0$ ($\ell = 1$) and $j \geq 3$ ($\ell \geq 5$). Only bands $j = -1, 1, 2$ remain, $1 + 48 + 192 = 241$ coefficients. A Fisher analysis of 30 days gives them 70%, 29% and 1.4% of the information on the sky, and it constrains only about 5 independent sky patterns.

Why not harmonics or pixels directly? Fifteen harmonics would describe what the data see, but each covers the whole sky, so a prior that says "the excess is compact" cannot be written in them. Pixels allow such a prior, but most pixel combinations are invisible and the visible ones are not separated from the rest. Needlets do both: their bands separate visible from invisible scales exactly, and within a band they are local, so a patch is described by a few nearby coefficients and the prior can be set per band (for instance $\beta_{jk} \sim \mathcal{N}(0, s_j^2)$ with s_j the spread of the smooth Galaxy's own coefficients in that band). Over a year, LISA's orientation sweeps through 360°, the response mixes the visible modes differently with time, and this band-by-band bookkeeping is what lets the model grow with the data.

25 Simulating a Galaxy with patches of excess power

One spectrum, many places. All Galactic power comes from unresolved double white dwarfs with the common spectrum $S^{\mathrm{GF}}(f)$ of Equation (5). The components of the sky differ only in *where* the power sits. The sky is a HEALPix map $I(\hat{n})$ of flux per pixel ($n_{\mathrm{side}} = 64$, 0.9° pixels), and the modulation is linear in it:

$$P_j(t) = \frac{\kappa}{\bar{R}} \sum_{\mathrm{pix}} I(\mathrm{pix})\, R_j(t, \mathrm{pix}), \qquad R_j = F^2_{+,j} + F^2_{\times,j}. \tag{67}$$

R_j is computed from first principles: arm unit vectors from the `bahamas` spacecraft orbit, Michelson responses X, Y, Z, and then $A = (Z - X)/\sqrt{2}$, $E = (X - 2Y + Z)/\sqrt{6}$. $\bar{R}$ is the all-sky mean of R_A, so an isotropic sky of unit flux gives $P_A = P_E = 1$. Because R_j has only $\ell \leq 4$, the map can be degraded to $n_{\mathrm{side}} = 32$ for this sum without any loss.

The smooth Galaxy is run 2's: the bivariate Gaussian of Section 4 in ecliptic (β, λ) with the pack-2 sky parameters, written out pixel by pixel. Since the envelope formula integrates $p(\beta, \lambda)$ with the measure $\cos\beta \, \mathrm{d}\beta \, \mathrm{d}\lambda$, the flux per pixel is $p(\beta, \lambda)\, \Delta\Omega$, normalised to a total of 1. The constant κ is fitted once so that this map reproduces run 2's analytic $M_j^2(t)$; it does so to 0.1% (A) and 0.2% (E) over the 30 days. The smooth part of the run-5 sky is therefore exactly run 2's injection, at the same amplitude.

Patches. Each patch i is an elliptical Gaussian on the sphere centred on galactic (l_i, b_i). For a pixel at angular distance θ from the centre and position angle φ (from galactic north towards increasing l),

$$I_i(\hat{n}) = F_i \, \frac{G_i(\hat{n})}{\sum_{\mathrm{pix}} G_i}, \qquad G_i = \exp\left[-\frac{1}{2} \left(\frac{\theta \cos(\varphi - \varphi_i)}{\sigma_{\mathrm{maj},i}} \right)^2 - \frac{1}{2} \left(\frac{\theta \sin(\varphi - \varphi_i)}{\sigma_{\mathrm{min},i}} \right)^2 \right], \tag{68}$$

with widths σ_{maj}, σ_{min} along and across the major axis, orientation φ_i, and F_i the patch's flux in units of the smooth Galaxy's. The full sky is $I = I_{\mathrm{gal}} + \sum_i I_i$, and by Equation (67)

$$P_j(t) = T_{j,\mathrm{gal}}(t) + \sum_i F_i \, T_{j,i}(t), \tag{69}$$

where $T_{j,i}$ is the modulation produced by patch i per unit flux. The time series is then simulated exactly as in Equation (36), $x_j = n_j + \sqrt{P_j(t)}\, g_j$, with the same seed as run 2: with the patches switched off, the series equals run 2's to 3×10^{-5}.

We place four patches where old stars, and hence white-dwarf binaries, are known to concentrate:

patch	(l, b)	$\sigma_{\mathrm{maj}} \times \sigma_{\mathrm{min}}$	F_i	why
LMC	$(280.5°, -32.9°)$	$3.0° \times 2.2°$	0.20	largest satellite galaxy
SMC	$(302.8°, -44.3°)$	$1.6° \times 1.0°$	0.08	20° from the LMC: two patches that blend at low ℓ
Sgr dSph	$(5.6°, -14.2°)$	$4.0° \times 1.5°$	0.12	on top of the smooth Galaxy, elongated across the plane
bar end	$(27°, 0°)$	$6.0° \times 1.5°$	0.12	in-plane overdensity that the Gaussian lacks

The fluxes are deliberately exaggerated. With the same number of binaries per stellar mass, the realistic flux fraction is M/d^2 relative to the Milky Way: about 6×10^{-4} for the LMC and below 10^{-4} for the SMC and Sgr, hundreds to thousands of times below what 30 days can detect. Over the 30 days the patches add 85% (A) and 56% (E) to the mean Galactic power; the largest single contribution is the LMC's (+41% in A, +28% in E). Each patch sits at a different place in the antenna pattern, so it adds a different, time-dependent amount to A and to E. That difference is the only handle the data have on where a patch is.

26 Run 5a: patch fluxes at known positions, and the role of needlets

The plan for run 5 has three steps of increasing freedom: (a) fit the patch fluxes with positions and shapes known; (b) free the positions; (c) replace the patches by needlet coefficients in bands $j = -1, 1, 2$ with a Gaussian prior per band. Only (a) has been run so far.

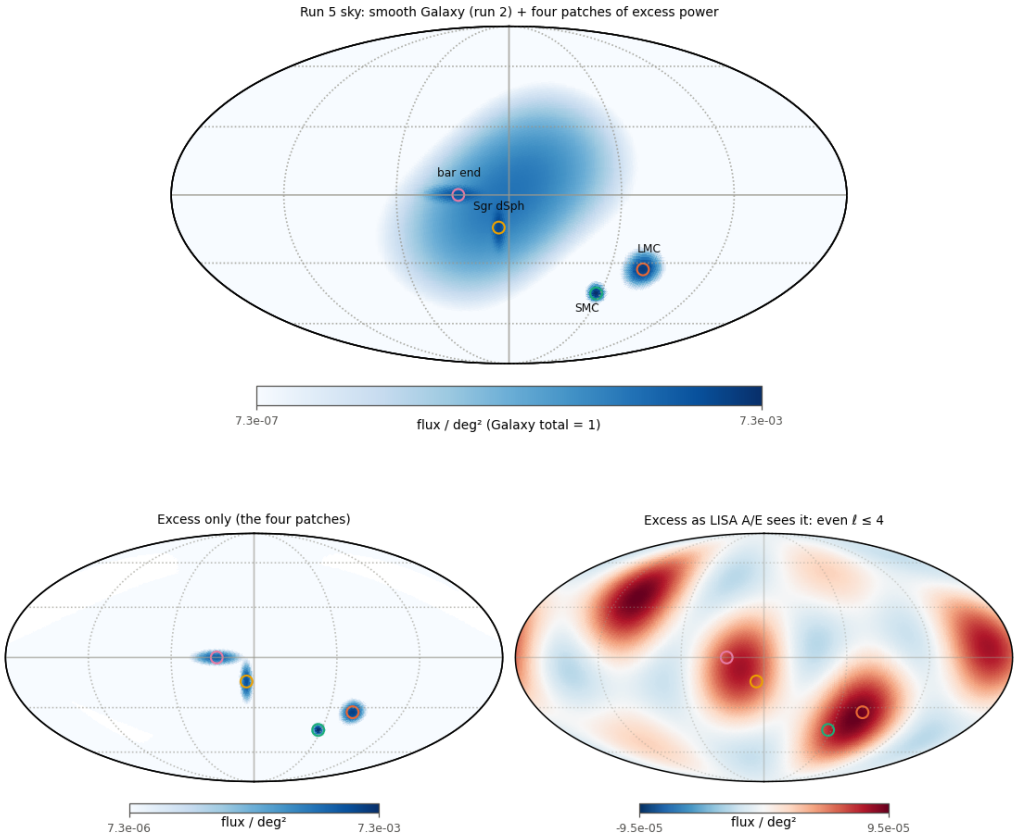


Figure 4: The run-5 sky in galactic coordinates. Top: smooth Galaxy (run 2's Gaussian) plus the four patches, log scale. Bottom left: the patches alone. Bottom right: the part of the patches LISA's A/E auto-power can see, their projection onto $\ell = 0, 2, 4$; it holds only 3.5% of the excess map's power, spread into lobes tens of degrees wide, and mirrored at the antipodes because the response is even.

The 5a model. The modulation is Equation (69), with $T_{j,\mathrm{gal}}$ fixed at the truth and the patch templates $T_{j,i}$ at their true positions and shapes. The free parameters are the seven spectral parameters and the four fluxes $F_i \sim U(0,1)$. The data, the WDM pixels and the Gamma likelihood are those of runs 1–4; only the model of $P_j(t)$ changes. This is the *best case* for any sky model: everything about the sky is known except how bright each patch is. Because the four templates are strongly correlated (below), the posterior is sampled with NUTS in whitened coordinates: each parameter is mapped from its uniform prior range to the real line with a logit, and then rotated and scaled with the curvature of the log-posterior at the truth. Plain NUTS reached its maximum tree depth.

What the needlets explain. Run 5a does not sample needlets, but the needlet picture of Section 24.4 is what explains its result and sets its place in the plan.

- *Only $\ell \leq 4$ is seen.* A 3° patch enters the data only through its $\ell = 0, 2, 4$ projection (Figure 4, bottom right), and every $T_{j,i}(t)$ is a combination of the same 15 harmonic responses. Over 30 days these change little, and each template is a smooth, nearly quadratic curve. The templates of different patches, and of the Galaxy, overlap by 79–99% (cosine similarity of the stacked A and E curves). The data can therefore measure the total excess well but not how it is split between patches.

- *5a is a constrained needlet model.* Each patch map has needlet coefficients $\beta_{jk}^{(i)}$, so Equa-

tion (69) is Equation (66) with $\beta = \beta^{\mathrm{gal}} + \sum_i F_i \beta^{(i)}$: the 241 visible coefficients restricted to move along four known directions. Run 5c will free all 241. It cannot do better than 5a on the fluxes, so 5a is the benchmark it will be compared with.

- *Positions are hard.* In the low-frequency limit $R_A + R_E$ depends only on the angle between $\hat{n}$ and the normal to the constellation plane, which turns by only about 30° in 30 days. A noise-free scan shows that a patch can be moved to any position at the same angle from the normal (a ring on the sky) at almost no cost. This is what run 5b will meet, and why a year of data, over which the normal precesses through 360°, is needed to localise anything.

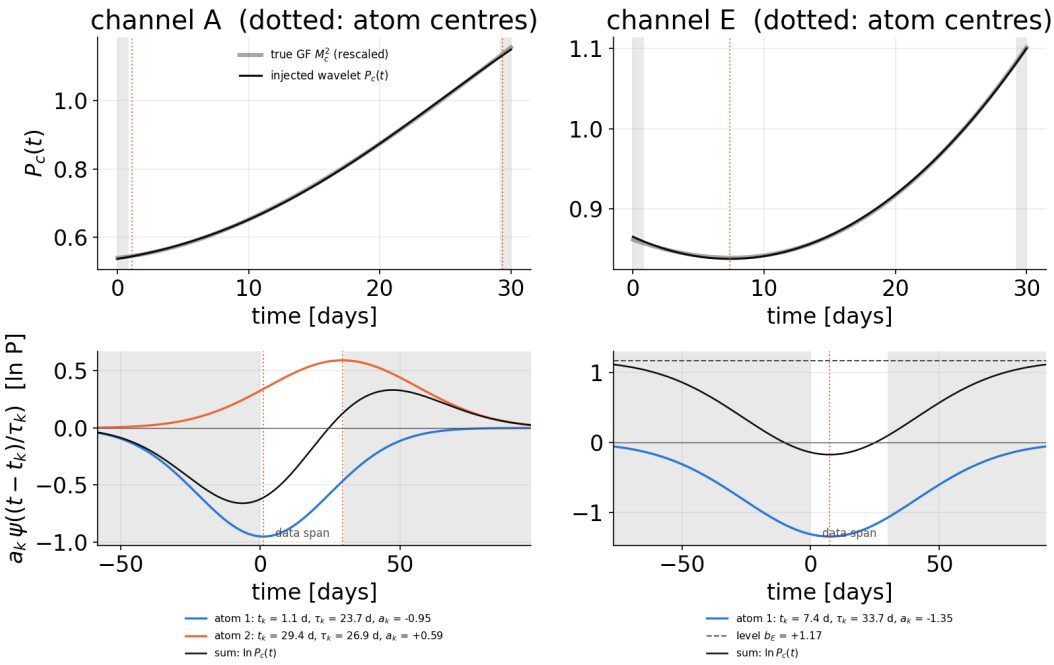


Figure 5: Run 1 injection. Top: the injected $P_j(t)$ built from atoms (black) on top of the true envelope (grey); grey bands are the dropped edge columns. Bottom: the individual atoms $a_k\psi((t-t_k)/\tau_k)$ in $\ln P$ and their sum, over a range wider than the data span (white) to show that the atoms extend beyond it.

Part V
Results

All runs use 30.00 days of simulated A and E data at $\mathrm{d}t = 10\,\mathrm{s}$, TDI generation 1, seed 2026, simulated with two extra days at each end and cut (non-periodic ends). The WDM grid has $N_f = 3600$ (72 pixels of 10 h); the likelihood drops two edge columns at each end and uses 34 blocks of 20 h × 174 log-frequency groups in 0.1–29 mHz. The atom runs use `eryn` with 32 walkers × 4 temperatures, 2000 burn-in and 6000 stored steps, starting from $K_j = 0$ (Section 19). Every run is converged: autocorrelation times are at most 8 steps (ESS ≥ 24,000) for all spectral parameters, K_j, $\log\mathcal{L}$ and $P_j(t)$ at several days.

The atom model's `amp` absorbs the level b_A and trades against the spectral shape, so it is not compared directly. The modulation is compared through the *galactic brightness*: for each posterior sample, its amplitude times its spectral shape (averaged over frequency with the weight of where the Galaxy dominates) times $P_j(t)$, divided by the injected one. This is the quantity the data constrain.

27 Run 1: recovering a modulation made of atoms

Set-up. The injected modulation is the least-squares atom fit to the true envelope (Section 13): two atoms in A, one in E, each wider than 23 d (Figure 5). The model can therefore represent the injection exactly.

Results. All eight spectral parameters are recovered within their 90% intervals, with every $|z| \le 1.3$ (Figure 6). Starting from $K_j = 0$, every walker finds the modulation within about 50

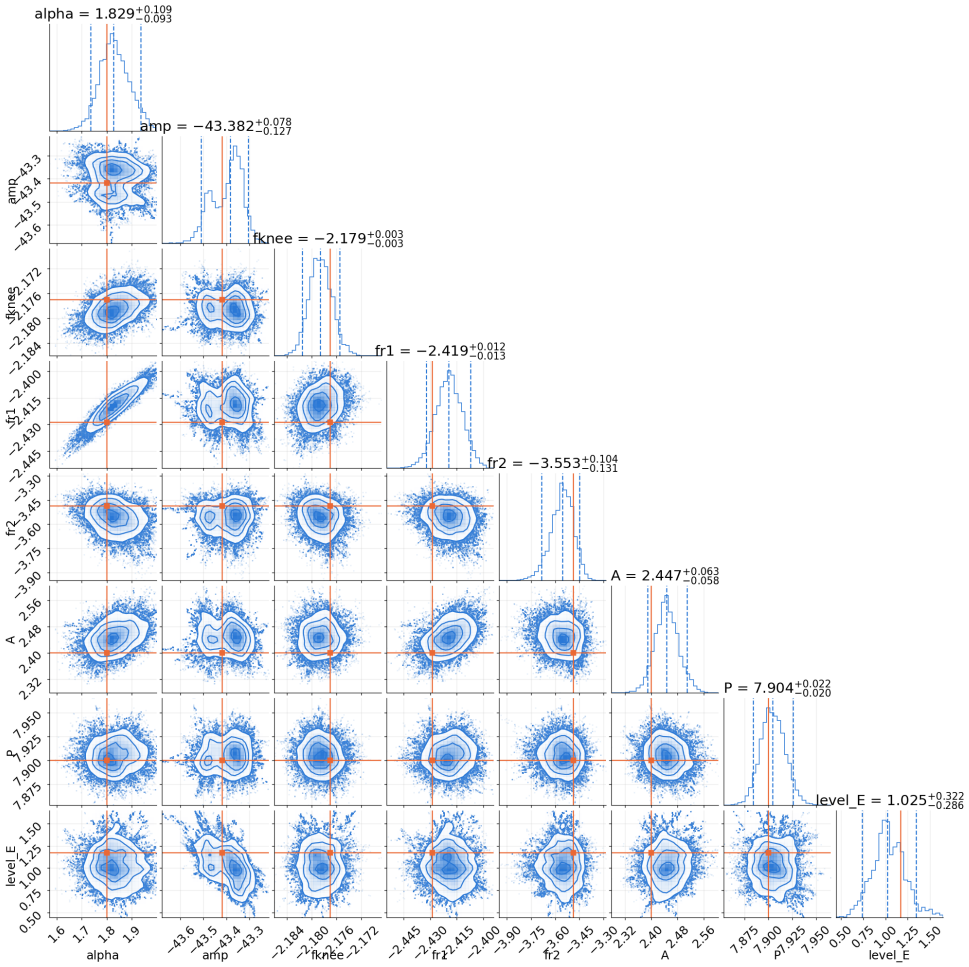


Figure 6: Run 1: posterior of the spectral parameters and `level_E`; orange lines are the injected values. The double peak in `amp` is the trade-off with the spectral shape and with the atoms.

steps, and $K_j = 0$ is never visited afterwards: the modulation is detected decisively. The number of atoms, however, is not identifiable: $p(K_A = 1) = 0.43$ against $p(K_A = 2) = 0.45$, and similarly in E. Over 30 days the envelope is smooth, and one or two wide atoms draw the same curve. The single E atom is recovered (centre 7.4 d inside $[3.6, 10.6]$ d at 90%); the two A atoms individually are not. What matters, the galactic brightness, is recovered to within ±1.5% in both channels, with a 90% band 5.5–5.7% wide. The one excursion is A at day 0 (+8%), where the data would like an atom centre before the start of the window, which the prior forbids.

28 Run 2: the true envelope, wavelets against chunks

Set-up. The Galactic foreground is modulated by the true sky envelope $M_j^2(t)$, which no finite set of atoms represents exactly. The *same* series is analysed with the wavelet RJMCMC (free atoms) and with the chunked pipeline of Part I (two 15-day chunks with a 5% Tukey taper, 1000 log bins, the 5-parameter sky envelope, `nessai`). The chunked likelihood is run twice: with the paper's shape $n_b/2$ and with the correct n_b (Section 18.3).

Results.

- **Spectral parameters** (Figure 8): the three posteriors agree on the medians (same data), all $|z| \leq 1.35$. The wavelet widths are 1.00–1.13 times those of the corrected chunked run.

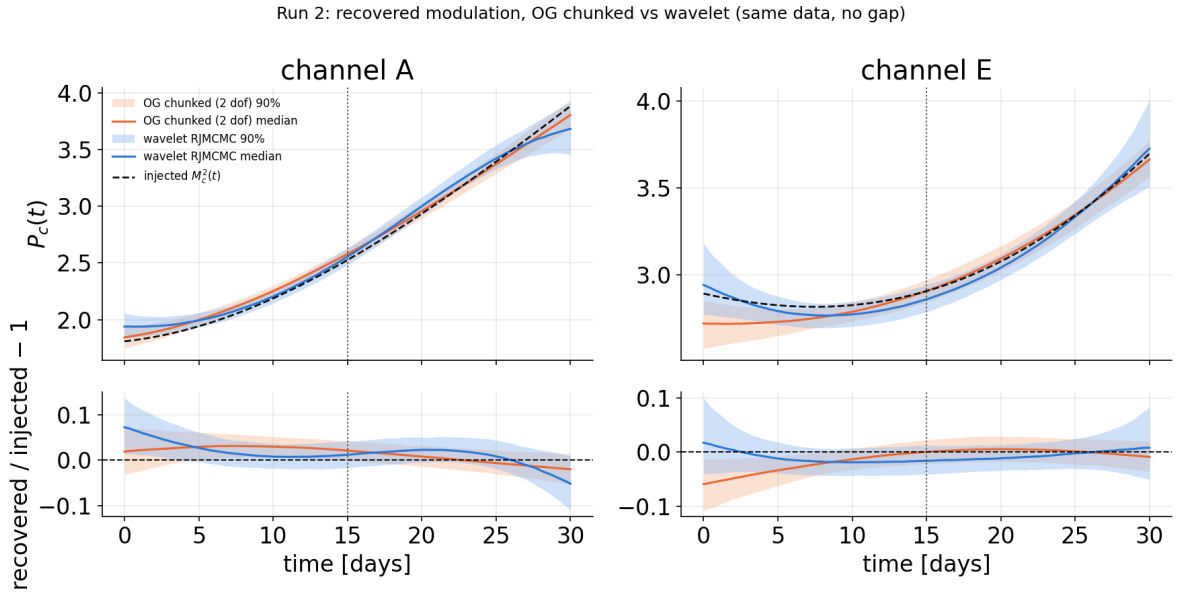


Figure 7: Run 2: recovered $P_j(t)$ on the injected scale, chunked pipeline with the correct degrees of freedom (orange) and wavelet RJMCMC (blue), 90% bands; dashed: the injected envelope. Bottom: recovered/injected −1. The dotted line marks the boundary between the two chunks.

With the paper's shape $n_b/2$ the chunked widths are 1.34–1.43 $\approx \sqrt{2}$ larger, as Section 18.3 predicts.

- **Galactic brightness** (Figure 7): 90% bands of 5.8% (A) and 5.2% (E) for the wavelet model, against 4.8% and 4.6% for the corrected chunked model, i.e. 12–20% wider. The difference sits mostly at the ends of the window, where the wavelet model has fewer pixels and the atoms extrapolate; in the middle the widths are close.
- **Time dependence alone** (each draw divided by its own 30-day mean): the wavelet bands are 1.3–1.5 times wider. The sky model ties both channels to five parameters, while free atoms fit each channel's shape on its own. This is the price of not assuming the Galaxy's shape.

The chunked sky parameters are all within 1.4σ of the truth but broad ($\sin\psi \in [-0.95, -0.52]$ at 90%): 30 days constrain the envelope's shape, not each sky parameter.

29 Runs 3 and 4: a one-day and a ten-day gap

Set-up. Run 2's series with a 1-day (run 3) or 10-day (run 4) gap, masked as in Section 22 (Figure 3). The sampler, priors and seeds are those of run 2, so the comparison with run 2's wavelet posterior isolates the effect of the gap.

Spectral parameters (Figure 9). The 1-day gap widens them by 0–8%, the 10-day gap by 11–32%. The expectation from the lost pixels, $\sqrt{1/(1-f_{\rm lost})}$, is 4% and 29%. All medians stay within 0.7σ of run 2's.

The modulation through the gap. As explained in Section 23, the atoms define $P_j(t)$ inside the gap without any data there.

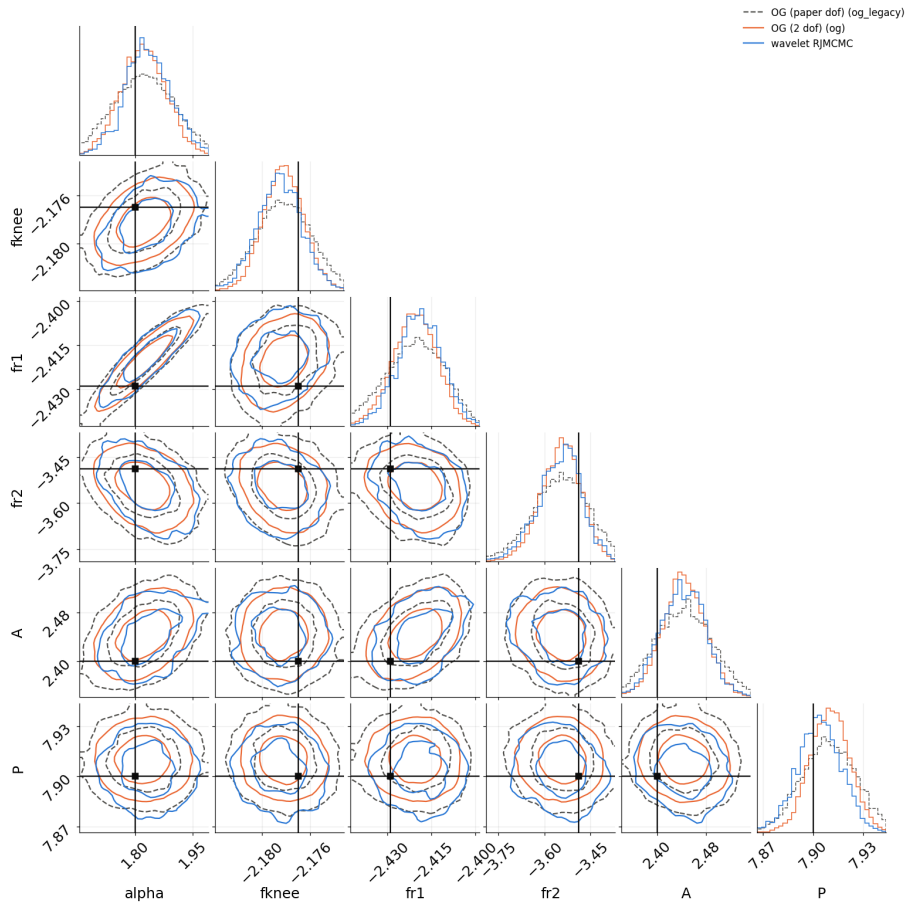


Figure 8: Run 2: spectral parameters shared by both models. Grey dashed: chunked pipeline with the paper's degrees of freedom; orange: chunked with the correct ones; blue: wavelet RJMCMC. Black lines: injected values.

- **1-day gap:** the band widens only locally, by 0.3–0.4 percentage points (from 5.9 to 6.2% in A, 5.4 to 5.8% in E), and the median follows run 2's within 0.5%. Atoms 20–35 d wide bridge 50 h with almost no extra freedom.
- **10-day gap** (Figure 10): the band grows steadily into the gap and peaks in its middle, at 14.5% (A) and 12.9% (E) against 5.8% and 5.3% without the gap, about 2.5 times wider. The median follows run 2's within 1%, and the truth stays inside the 90% band throughout the gap in both channels. The kept data widen slightly too (6.1 to 7.5% in A over 3–27 d), because the shared spectral parameters are less well measured.

Neither gap introduces a bias. A gap costs what the missing data account for, plus an interpolation uncertainty inside the gap that the posterior reports honestly.

30 Run 5: a Galaxy with four bright patches

Set-up. The sky of Section 25 (Figure 4), analysed on the same WDM pixels with two models of $P_j(t)$: the atoms in time of runs 1–4, and the 5a patch fit of Section 26.

The total modulation (Figure 11). Both models recover the 30-day mean galactic brightness to within 1.5% in A and 1.2% in E, with the same 90% width (3.2%). The spectral parameters agree, with width ratios 5a/atoms of 0.88–1.06. Along time, the 5a band is somewhat narrower

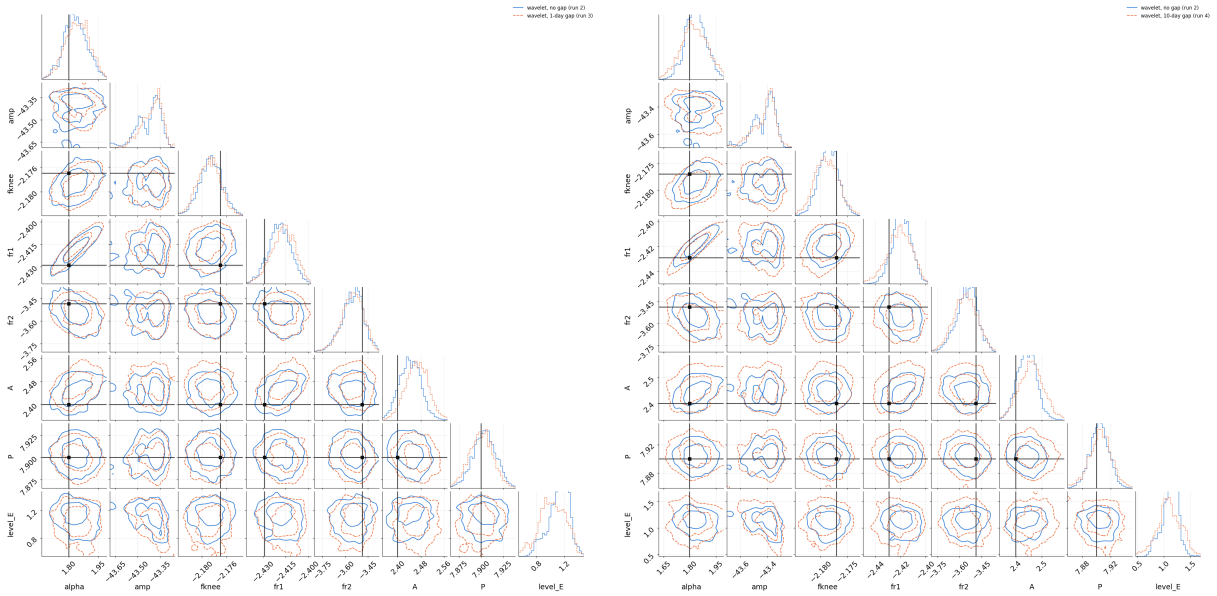


Figure 9: Spectral posteriors with a gap (orange dashed) against run 2 without a gap (blue). Left: run 3, 1-day gap. Right: run 4, 10-day gap. Black lines: injected values.

(4.2–4.3% against 4.9–5.6%), as expected from a model that knows the sky. Even a large departure from the Gaussian Galaxy (+85% of power in A) is therefore absorbed by the atoms without bias: they describe $P_j(t)$, whatever its origin.

The individual patches. Detection significance is the posterior median over its standard deviation:

patch	true F_i	median	90% interval	median/std	(median−truth)/std
LMC	0.20	0.133	[0.013, 0.331]	1.4	−0.69
SMC	0.08	0.118	[0.047, 0.181]	2.9	+0.92
Sgr dSph	0.12	0.227	[0.055, 0.434]	1.9	+0.92
bar end	0.12	0.200	[0.122, 0.274]	4.3	+1.71

All four fluxes are consistent with the truth, but only the bar end, the one patch in the plane and away from the centre, is clearly detected. The fluxes are strongly anticorrelated (LMC with the others, −0.6 to −0.75). The reason is the template overlap of Section 26: alone, each flux would be measured to 0.02–0.07; with all four free, the errors grow two to four times. The data measure about two flux combinations well (errors 0.020 and 0.025) and the other two poorly (0.07 and 0.25). *Even in the best case, 30 days measure how much excess there is, not where it comes from.*

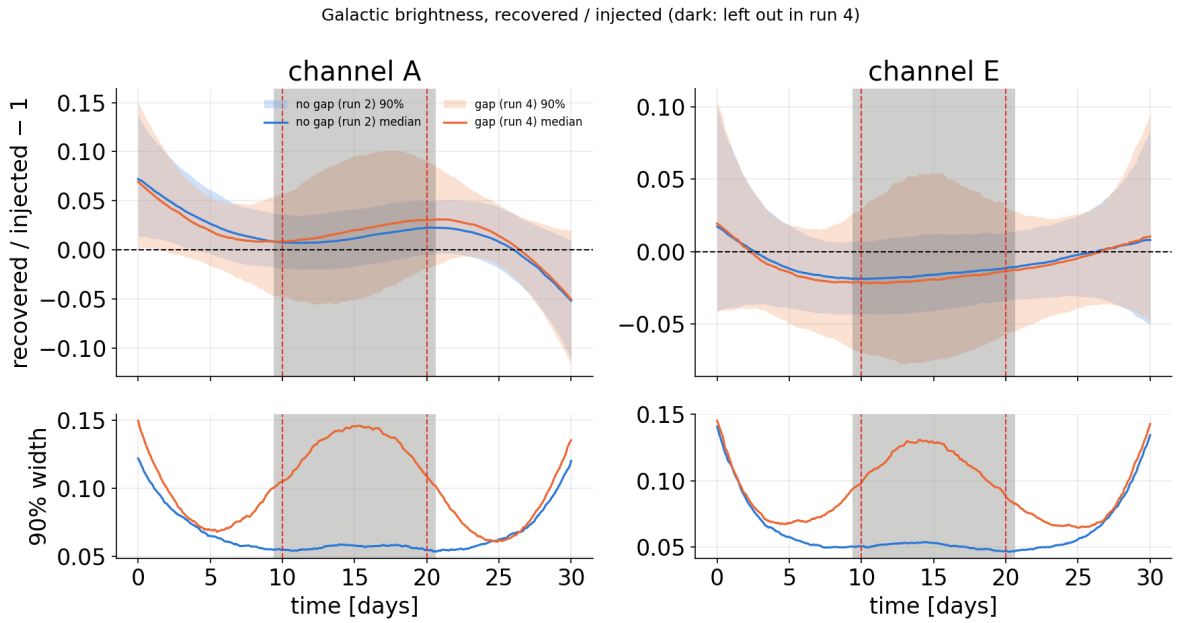


Figure 10: Run 4: galactic brightness recovered/injected −1 with the 10-day gap (orange) and without it (run 2, blue). Grey: the columns left out of the likelihood; red dashed: the gap. Inside the grey region the curve is the atom model continued through the gap, constrained only by the data on either side. Bottom: width of the 90% band.

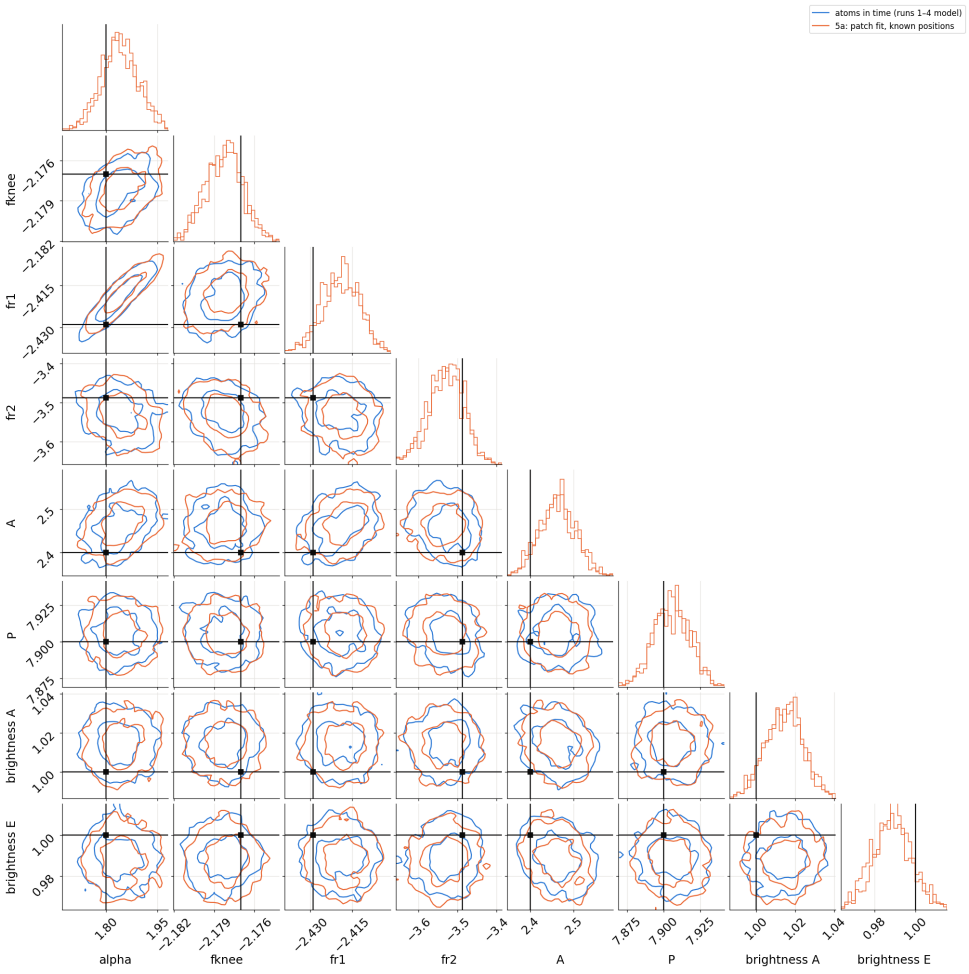


Figure 11: Run 5: atoms in time (blue) and the 5a patch fit at known positions (orange) on the same data, for the shared spectral parameters and the 30-day mean galactic brightness in A and E (recovered/injected). Black lines: injected values.

Part VI
Conclusions and outlook

31 Conclusions

- **The WDM likelihood is the chunked likelihood in another domain.** The Gamma likelihood of grouped WDM pixels is Buscicchio, Pozzoli et al. (2025), Eq. (9) applied to wavelet periodograms, with 20-h time blocks in place of chunks. The derivation shows that Buscicchio, Pozzoli et al. (2025), Eq. (9) (and the chunked samplers of `bahamas`) uses half the correct degrees of freedom, which makes chunked posteriors $\sqrt{2}$ too wide.
- **A free modulation is recovered without bias.** Gaussian atoms with a variable number, sampled by reversible jump, recover the galactic brightness to 1–2% with 5–6% bands over 30 days, both for an atom injection and for the true sky envelope. The number of atoms is not identifiable, and need not be.
- **Its price is modest.** Against the 5-parameter sky model on the same data, the wavelet brightness bands are 12–20% wider and the time dependence alone 1.3–1.5 times wider, mostly at the ends of the window. The spectral parameters are measured equally well.
- **Gaps are handled exactly by masking.** Leaving out the gap columns and one buffer column per side keeps the likelihood exact. A 1-day gap costs almost nothing; a 10-day gap widens the spectral parameters as the lost data predict and the modulation band inside the gap by 2.5 times, without bias.
- **The sky is seen only through $\ell = 0, 2, 4$.** A patchy Galaxy changes $P_j(t)$ strongly, and the atoms absorb it without bias. Even with positions known, the fluxes of individual patches are barely separable in 30 days. Needlets make the reason explicit and reduce a sky model to 241 coefficients in three bands.

32 Discussion and next steps

From 30 days to four years. Everything above uses 30 days, over which the envelope is a smooth, nearly quadratic curve and the constellation normal turns by only about 30°. The next step is to apply the pipeline to four years of mock LISA data. Over that span the modulation goes through eight cycles, the atom model must describe a periodic signal with many more atoms, and the sky model gains the most: as the normal precesses through 360°, the ring degeneracy of patch positions should break, and the needlet coefficients in bands $j = 1, 2$ should become measurable. The likelihood scales linearly with the number of time blocks and should remain cheap; we expect the transdimensional sampling to be the bottleneck. Runs 5b (free positions) and 5c (needlet coefficients) will be done on both time scales.

Better handling of gaps. Runs 3 and 4 place one hard-edged gap in the middle of the window. Open tests are: the realistic LISA gap pattern (short scheduled gaps every two weeks plus random unscheduled ones, a 75% duty cycle); a long gap at the start or end of the window, where the atoms extrapolate instead of interpolate; tapered gap edges, which should remove the low-frequency leakage and shrink the buffer; letting atom centres lie outside the window, which should cure the excess at the window ends seen in every run; and the chunked pipeline on the same gapped data, where a short gap can cost most of a chunk.

Comparison with CTMCMC. The number of atoms (and later of sky components) is sampled here with birth–death reversible-jump moves in `eryn`: a birth or death is proposed at discrete steps and accepted or rejected with a Metropolis–Hastings probability. `samsara` (Astorino et al. 2025) is a continuous-time MCMC (CTMCMC) sampler built for the same kind of transdimensional problem, with the LISA global fit as its motivation. Following the birth–death process of Stephens (2000), components are born, die and mutate as Poisson events in continuous time. The rates are chosen so that detailed balance holds, and every transdimensional move is then accepted; the posterior is obtained by weighting each state by the time the chain spends in it. This removes the low acceptance of birth and death moves (0.7–1% in our runs), which is the main limit on how fast the reversible-jump chain explores K. We plan to fit the atom model, and later the needlet sky model, with both samplers on the same data and compare the posteriors on the modulation, the mixing in K, and the cost per independent sample.

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